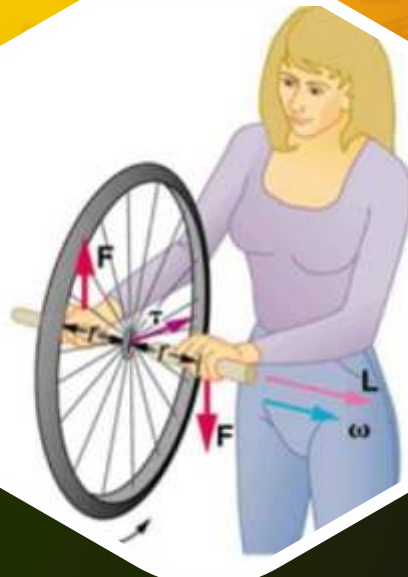


PHYSICS

JEE (MAIN+ADVANCED)

NURTURE COURSE



Study Material

Rotational Dynamics

(English Medium)

Copyright Statement

All rights including trademark and copyrights and rights of translation etc. reserved and vested exclusively with ALLEN Career Institute Private Limited. (ALLEN)

No part of this work may be copied, reproduced, adapted, abridged or translated, transcribed, transmitted, stored or distributed in any form retrieval system, computer system, photographic or other system or transmitted in any form or by any means whether electronic, magnetic, chemical or manual, mechanical, digital, optical, photocopying, recording or otherwise, or stored in any retrieval system of any nature without the written permission of the Allen Career Institute Private Limited. Any breach will entail legal action and prosecution without further notice.

This work is sold/distributed by Allen Career Institute Private Limited subject to the condition and undertaking given by the student that all proprietary rights (under the Trademark Act, 1999 and Copyright Act, 1957) of the work shall be exclusively belong to ALLEN Career Institute Private Limited. Neither the Study Materials and/or Test Series and/or the contents nor any part thereof i.e. work shall be reproduced, modify, re-publish, sub-license, upload on website, broadcast, post, transmit, disseminate, distribute, sell in market, stored in a retrieval system or transmitted in any form or by any means for reproducing or making multiple copies of it.

Any person who does any unauthorised act in relation to this work may be liable to criminal prosecution and civil claims for damages. Any violation or infringement of the propriety rights of Allen shall be punishable under Section- 29 & 52 of the Trademark Act, 1999 and under Section- 51, 58 & 63 of the Copyright Act, 1957 and any other Act applicable in India. All disputes are subjected to the exclusive jurisdiction of courts, tribunals and forums at Kota, Rajasthan only.

Note:- This publication is meant for educational and learning purposes. All reasonable care and diligence have been taken while editing and printing this publication. ALLEN Career Institute Private Limited shall not hold any responsibility for any error that may have inadvertently crept in. ALLEN Career Institute Private Limited is not responsible for the consequences of any action taken on the basis of this publication.

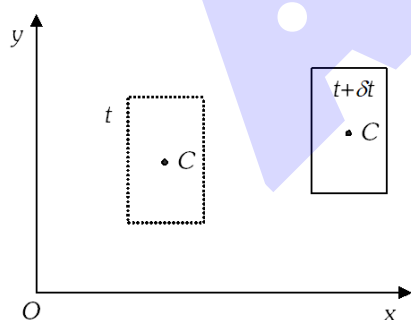
ROTATIONAL DYNAMICS

Till so far we have learnt kinematics and kinetics of translation motion in which all the particles of a body undergo identical motions i.e. at any instant of time all of them have equal velocities and equal accelerations and in any interval of time they all follow identical trajectories. Therefore kinematics of any particle of a body or of its mass center in translation motion is representative of kinematics of the whole body. But when a body is in rotation motion, all of its particles and the mass center do not undergo identical motions. Newton's laws of motion, which are the main guiding laws of mechanics, are applicable to a point particle and if applied to a rigid body or system of particles, they predict motion of the mass center. Therefore, it becomes necessary to investigate how mass center and different particles of a rigid body move when the body rotates. In kinematics of rotation motion we investigate relations existing between time, positions, velocities and accelerations of different particles and mass center of a rigid body in rotation motion.

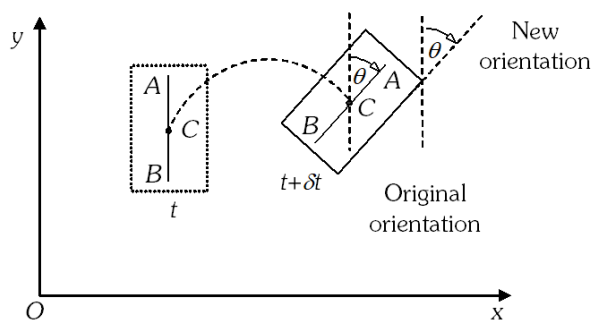
1.0 RIGID BODY

A rigid body is an assemblage of a large number of material particles, which do not change their mutual distances under any circumstance or in other words, they are not deformed under any circumstance.

Actual material bodies are never perfectly rigid and are deformed under action of external forces. When these deformations are small enough to be considered during their course of motion, the body is assumed a rigid body. Hence, all solid objects such as stone, ball, vehicles etc are considered as rigid bodies while analysing their translation as well as rotation motion.



Pure Translation



Combination of translation and rotation

To analyse rotation of a body, relative motion between its particles cannot be neglected and size of the body becomes a considerable factor. This is why study of rotation motion is also known as mechanics of rigid bodies.

1.1 Rotation Motion of a Rigid Body

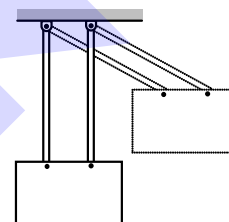
Any kind of motion of a body is identified by change in position or change in orientation or change in both. If a body changes its orientation during its motion it said to be in rotation motion. In the following figures, a rectangular plate is shown moving in the x-y plane. The point C is its mass center. In the first case it does not changes orientation, therefore is in pure translation motion. In the second case it changes its orientation during its motion. It is a combination of translation and rotation motion.

Rotation i.e. change in orientation is identified by the angle through which a linear dimension or a straight line drawn on the body turns. In the figure this angle is shown by θ .

Illustrations

Illustration 1. Identify Translation and rotation motion

A rectangular plate is suspended from the ceiling by two parallel rods each pivoted at one end on the plate and at the other end on the ceiling. The plate is given a side-push to oscillate in the vertical plane containing the plate. Identify motion of the plate and the rods.



Solution.

Neither of the linear dimensions of the plate turns during the motion. Therefore, the plate does not change its orientation. Here edges of the body easily fulfill our purpose to measure orientation; therefore, no line is drawn on it.

The plate is in curvilinear translation motion and the rods are in rotation motion.

1.2 Types of Motions involving Rotation

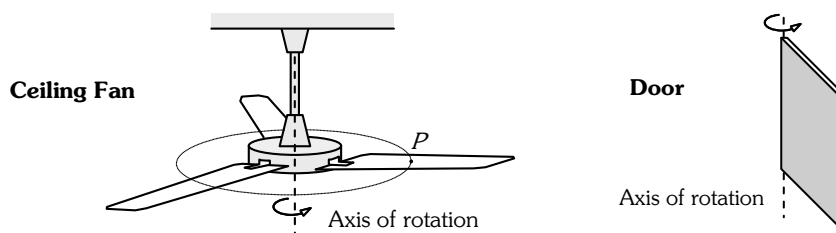
Motion of body involving rotation can be classified into following three categories.

- I Rotation about a fixed axis.
- II Rotation about an axis in translation.
- III Rotation about an axis in rotation

Rotation about a fixed axis

Rotation of ceiling fan, potter's wheel, opening and closing of doors and needles of a wall clock etc. come into this category.

When a ceiling fan it rotates, the vertical rod supporting it remains stationary and all the particles on the fan move on circular paths. Circular path of a particle P on one of its blades is shown by dotted circle. Centers of circular paths followed by every particle are on the central line through the rod. This central line is known as axis of rotation and is shown by a dashed line. All the particles on the axis of rotation are at rest, therefore the axis is stationary and the fan is in rotation about this fixed axis.

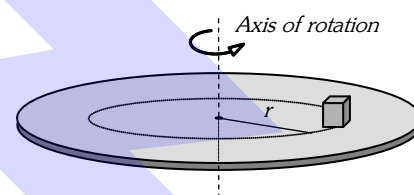


A door rotates about a vertical line that passes through its hinges. This vertical line is the axis of rotation. In the figure, the axis of rotation is shown by dashed line.

Axis of rotation

An imaginary line perpendicular to plane of circular paths of particles of a rigid body in rotation and containing the centers of all these circular paths is known as axis of rotation.

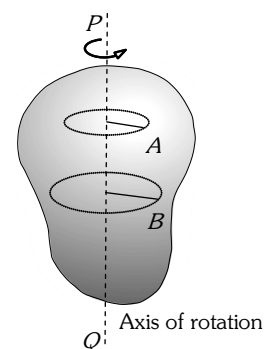
It is not necessary that the axis of rotation pass through the body. Consider system shown in the figure, where a block is fixed on a rotating disc. The axis of rotation passes through the center of the disc but not through the block.



Important observations

Let us consider a rigid body of arbitrary shape rotating about a fixed axis PQ passing through the body. Two of its particles A and B shown are moving on their circular paths.

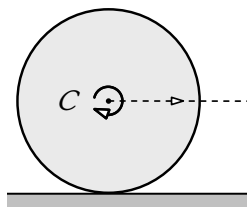
- All of its particles, not on the axis of rotation, move on circular paths with centers on the axis of rotation. All these circular paths are in parallel planes that are perpendicular to the axis of rotation.
- All the particles of the body cover same angular displacement in the same time interval, therefore all of them move with the same angular velocity and angular acceleration.
- Particles moving on circular paths of different radii move with different speeds and different magnitudes of linear acceleration. Furthermore, no two particles in the same plane perpendicular to the axis of rotation have same velocity and acceleration vectors.
- All the particles on a line parallel to the axis of rotation move circular paths of the same radius therefore have same velocity and acceleration vectors.
- Consider two particles in a plane perpendicular to the rotational axis. Every such particle on a rigid body in rotation motion moves on circular path relative to another one. Radius of the circular path equals to the distance between the particles. In addition, angular velocity and angular acceleration equals to that of rotation motion of the body.



Rotation about an axis in translation

Rotation about an axis in translation includes a broad category of motions. Rolling is an illustration of this kind of motion. A rod lying on table when pushed from its one end in its perpendicular direction also executes this kind of motion. To understand more let us discuss few illustrations.

Consider rolling of wheels of a vehicle, moving on straight level road. Relative to a reference frame, moving with the vehicle wheel appears rotating about its stationary axel. The rotation of the wheel from this frame is rotation about fixed axis. Relative to a reference frame fixed with the ground, the wheel appears rotating about the moving axel, therefore, rolling of a wheel is superposition of two simultaneous but distinct motions – rotation about the axel fixed with the vehicle and translation of the axel together with the vehicle.

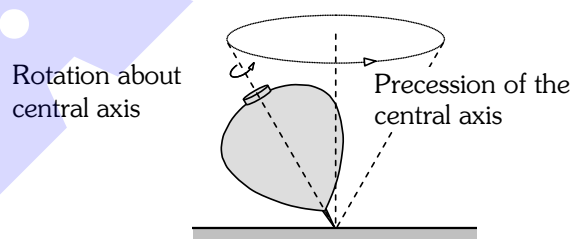


Important observations

- Every particle of the body always remains in a plane perpendicular to the rotational axis. Therefore, this kind of motion is also known as general plane motion.
- Relative to every particle another particle in a plane perpendicular to axis of rotation moves on circular path. Radius of the circular path equals to the distance between the particles, angular velocity and angular acceleration equals to that of rotation motion of the body.
- Rotation about axis in translation is superposition of pure rotation about the axis and simultaneous translation motion of the axis.

Rotation about an axis in rotation

In this kind of motion, the body rotates about an axis that also rotates about some other axis. Analysis of rotation about rotating axes is not in the scope of JEE, therefore we will discuss it to have an elementary idea only.



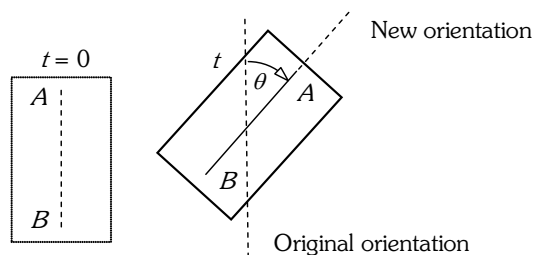
As an illustration consider a rotating top. The top rotates about its central axis of symmetry and this axis sweeps a cone about a vertical axis. The central axis continuously changes its orientation, therefore is in rotation motion. This type of rotation in which the axis of rotation also rotates and sweeps out a cone is known as precession.

Another illustration of rotation about axis in rotation is a table-fan swinging while rotating. Table-fan rotates about its horizontal shaft along which axis of rotation passes. When the rotating table-fan swings, its shaft rotates about a vertical axis.

1.3 Angular displacement, angular velocity and angular acceleration

Rotation motion is the change in orientation of a rigid body with time. It is measured by turning of a linear dimension or a straight line drawn on the body.

In the figure is shown at two different instants $t = 0$ and t a rectangular plate moving in its own plane. Change in orientation during time t equals to the angle θ through which all the linear dimensions of the plate or a line AB turns.



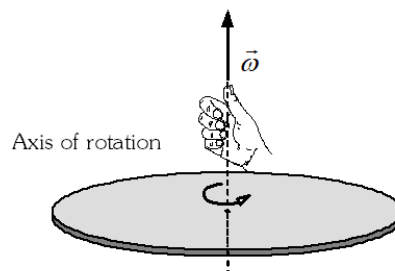
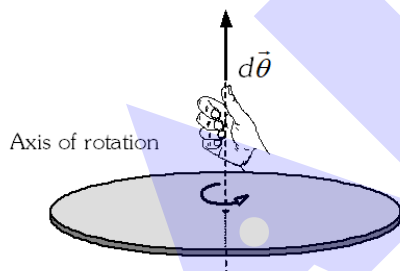
If the angle θ continuously changes with time t , instantaneous angular velocity ω and angular acceleration α for rotation of the body are defined by the following equations.

$$\omega = \frac{d\theta}{dt} \quad [1]$$

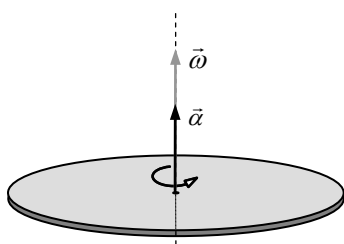
$$\alpha = \frac{d^2\theta}{dt^2} = \frac{d\omega}{dt} = \omega \frac{d\omega}{d\theta} \quad [2]$$

Direction of angular motion quantities

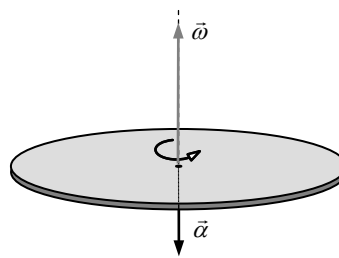
Angular displacement, angular velocity and angular acceleration are known as angular motion quantities. Infinitesimally small angular displacement, instantaneous angular velocity and angular acceleration are vector quantities. Direction of infinitesimally small angular displacement and instantaneous angular velocity is given by the right hand rule. For a disc rotating as shown in the figure, the angular velocity points upwards along the axis of rotation.



The direction of angular acceleration depends on whether angular velocity increases or decreases with time. For increasing angular velocity, the angular acceleration vector points in the direction of angular velocity vector and for decreasing angular velocity, the angular acceleration vector points opposite to the angular velocity vector.



Angular acceleration: Increasing angular speed



Angular acceleration: Decreasing angular speed

In rotation about fixed axis and rotation about axis in translation, the axis of rotation does not rotate and angular velocity and acceleration always point along the axis of rotation. Therefore, in dealing these kinds of motions, the angular motion quantities can be used in scalar notations by assigning them positive sign for one direction and negative sign for the opposite direction.

These quantities have similar mathematical relations as position coordinate, velocity, acceleration and time have in rectilinear motion.

- A body rotating with constant angular velocity ω and hence zero angular acceleration is said to be uniform rotation. Angular position θ is given by equation

$$\theta = \theta_0 + \omega t \quad [3]$$

- Thus for a body rotating with uniform angular acceleration α , the angular position θ and angular velocity ω can be expressed by the following equation.

$$\omega = \omega_0 + \alpha t \quad [4]$$

$$\theta = \theta_0 + \omega_0 t + \frac{1}{2} \alpha t^2 = \theta_0 + \frac{1}{2} (\omega_0 + \omega) t \quad [5]$$

$$\omega^2 = \omega_0^2 + 2\alpha(\theta - \theta_0) \quad [6]$$

Angular motion quantities in rotation and assumption of axis of rotation

Rotation is identified by change in orientation, which is measured by turning of a linear dimension of the body or a line drawn on the body. It remains unchanged relative to all inertial frames.

Therefore, if we assume axis of rotation anywhere but parallel to the original one, angular displacement, angular velocity and angular acceleration of rotation motion remain the same.

Illustrations

Illustration 2. A wheel is rotating with angular velocity 2 rad/s. It is subjected to uniform angular acceleration 2.0 rad/s².

- How much angular velocity does the wheel acquire after 10 s?
- How many complete revolution it makes in this time interval?

Solution. The wheel is in uniform angular acceleration,

$\omega = \omega_0 + \alpha t \rightarrow$ Substituting values of ω_0 , α and t , we have

$$\omega = 2 + 2 \times 10 = 22 \text{ rad/s}$$

$$\theta = \theta_0 + \frac{1}{2} (\omega_0 + \omega) t \rightarrow \text{Substituting } \theta_0 = 0$$

for initial position, and ω_0 from above equation, we have

$$\theta = 0 + \frac{1}{2} (2 + 22) 10 = 120 \text{ rad}$$

In one revolution, the wheel rotates through 2π radians. Therefore, number of complete revolutions n is

$$n = \frac{\theta}{2\pi} = \frac{60}{\pi} \text{ rev.}$$

Illustration 3. A disc rotates about a fixed axis. Its angular velocity ω varies with time according to equation $\omega = at + b$. At the instant $t = 0$ its angular velocity is 1.0 rad/s and at angular position is 2 rad and at the instant $t = 2$ s, angular velocity is 5.0 rad/s. Determine angular position θ and angular acceleration α when $t = 4$ s.

Solution The given equation $\omega = at + b$ has form similar to eq. [4], therefore motion is rotation with uniform angular acceleration.

Initial angular velocity = $\omega_0 = b = 1.0$ rad/s

Angular acceleration $\alpha = a$

Substituting these values in eq. [5], we get

$$\theta = \frac{1}{2}at^2 + \omega_0 t + c$$

Since at $t = 0$, $\omega = 1.0$ rad/s, we obtain the constant c .

Initial angular position = $\theta_0 = c = 2.0$ rad

Since at $t = 2.0$ s angular velocity is 5.0 rad/s, from given expression of angular velocity, we have

$\omega = at + b \rightarrow$ Substituting $b = 1.0$ rad/s, $t = 2.0$ s and $\omega = 5.0$ rad/s, we have
 $a = 2.0$ rad/s²

Now we can write expressions for angular position, angular velocity and angular acceleration.

$$\theta = t^2 + t + 2.0 \quad \dots(1)$$

$$\omega = 2.0t + 1.0 \quad \dots(2)$$

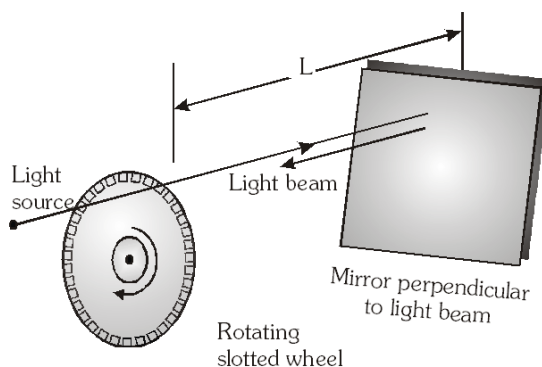
From the above equations, we can calculate angular position, angular velocity and angular acceleration at $t = 4.0$ s

$$\theta_4 = 22 \text{ rad}, \omega_4 = 9.0 \text{ rad/s}, \alpha = 2.0 \text{ rad/s}^2$$

Illustration 4. An early method of measuring the speed of light makes use of a rotating slotted wheel. A beam of light passes through slot at the outside edge of the wheel, as shown in figure below, travels to a distant mirror, and returns to the wheel just in time to pass through the next slot in the wheel. One such slotted wheel has a radius of 5.0 cm and 500 slots at its edge. Measurements taken when the mirror is $L = 500$ m from the wheel indicate a speed of light of 3.0×10^5 km/s.

(a) What is the (constant) angular speed of the wheel?

(b) What is the linear speed of a point on the edge of the wheel?



Solution

- (a) During the time light goes from the wheel to the mirror and comes back again, the wheel turns through an angle of $\theta = \frac{2\pi}{500} = 1.26 \times 10^{-2}$ rad.

$$\text{That time is } t = \frac{2L}{c} = \frac{2(500\text{m})}{2.998 \times 10^8 \text{ m/s}} = 3.34 \times 10^{-6} \text{ s}$$

$$\text{So the angular speed of the wheel is } \omega = \frac{\theta}{t} = \frac{1.26 \times 10^{-2} \text{ rad}}{3.34 \times 10^{-6} \text{ s}} = 3.8 \times 10^3 \text{ rad/s}$$

- (b) Linear speed of a point on the edge of a wheel

$$v = \omega r = 3.8 \times 10^3 \times 0.05 = 1.9 \times 10^2 \text{ m/s}$$

Illustration 5. A pulsar is rapidly rotating neutron star that emits a radio beam like a lighthouse emits a light beam. We receive a radio pulse for each rotation of the star. The period T of rotation is found by measuring the time between pulses. The pulsar in the Crab nebula has a period of rotation of $T = 0.033 \text{ s}$ that is increasing at the rate of 1.26×10^{-5} second/year.

- (a) What is the pulsar's angular acceleration?
 (b) If its angular acceleration is constant, how many years from now will the pulsar stop rotating?
 (c) The pulsar originated in a supernova explosion seen in the year 1054. What was the initial T for the pulsar? (Assume constant angular acceleration since the pulsar originated.).

Solution

- (a) The angular velocity in rad/s $\omega = 2\pi/T$.

$$\text{The angular acceleration} = \alpha = \frac{d\omega}{dt} = -\frac{2\pi}{T^2} \frac{dT}{dt}$$

$$\text{For the pulsar described } \frac{dT}{dt} = \frac{1.26 \times 10^{-5} \text{ s/y}}{3.16 \times 10^7 \text{ s/y}} = 4.00 \times 10^{-13}$$

$$\text{So } \alpha = -\left[\frac{2\pi}{(0.033\text{s})^2}\right](4.00 \times 10^{-13}) = -2.3 \times 10^{-3} \text{ rad/s}^2$$

The negative sign indicates that the angular acceleration is opposite the angular velocity and the pulsar is slowing down.

- (b) $\omega = \omega_0 + \alpha t$ for the time t when $\omega=0$.

$$t = -\frac{\omega}{\alpha} = -\frac{2\pi}{\alpha T} = -\frac{2\pi}{(-2.3 \times 10^{-3} \text{ rad/s}^2)(0.033\text{s})} = 8.3 \times 10^{10} \text{ s}.$$

This is about 2600 years.

- (c) The pulsar was born $1992 - 1054 = 938$ years ago.

This is equivalent to $(938 \text{ y})(3.16 \times 10^7 \text{ s/y}) = 2.96 \times 10^{10} \text{ s}$. Its angular velocity was then

$$\omega = \omega_0 + \alpha t = \frac{2\pi}{T} + \alpha t = \frac{2\pi}{0.033\text{s}} + (-2.3 \times 10^{-3} \text{ rad/s}^2)(-2.96 \times 10^{10} \text{ s}) = 258 \text{ rad/s}.$$

$$\text{Its period was } T = \frac{2\pi}{\omega} = 2.4 \times 10^{-2} \text{ s}.$$

Illustration 6. A turn table is rotating in a horizontal plane about the vertical axis passing through its centre with an angular velocity 20 rad/s. It carries upon it a flywheel rotating with an angular velocity 40 rad/s about a horizontal axle mounted in bearings. Find the angular velocity of the wheel as seen by an observer in the room.

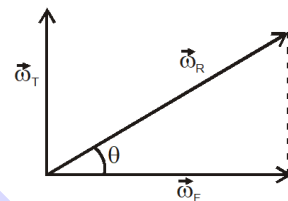
Solution As the axis of the turn table is vertical its angular velocity ω_T is directed vertical. The axis of flywheel is horizontal therefore its angular velocity ω_F is directed horizontal, hence the resultant angular velocity is

$$\vec{\omega}_R = \vec{\omega}_F + \vec{\omega}_T$$

$$|\vec{\omega}_R| = \sqrt{\omega_F^2 + \omega_T^2} = \sqrt{40^2 + 20^2} = 20\sqrt{5} \text{ rad/s}$$

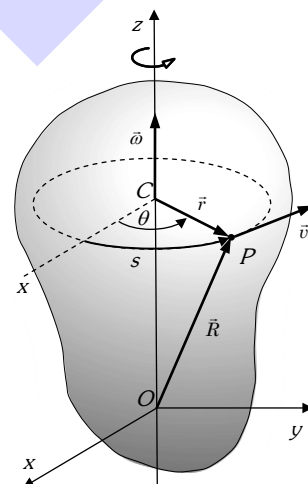
$\vec{\omega}_R$ lies in a plane which makes an angle θ with the horizontal

plane, given by $\theta = \tan^{-1} \left(\frac{\omega_T}{\omega_F} \right) = \tan^{-1} \left(\frac{1}{2} \right)$



1.4 Kinematics of rotation about fixed axis

In figure is shown a rigid body of arbitrary shape rotating about the z-axis. In the selected frame (here the coordinate system) all the three axes are at rest, therefore the z-axis that is the axis of rotation is at rest and the body is in fixed axis rotation. All of its particles other than those on the z-axis move on circular paths with their centers on the z-axis. All these circular paths are parallel to the x-y plane. In the figure, one of its particles P is shown moving with velocity $\vec{v}$ on a circular path of radius r and center C. Its position vector is $\vec{R}$. It were at the line Cx at $t = 0$ and at the position shown at the instant t . During time interval t , it covers the circular arc of length s and its radius vector turns through angle θ .



In an infinitesimally small time interval dt let, the particle covers infinitesimally small distance ds along its circular path.

$$d\vec{s} = d\vec{\theta} \times \vec{r} = d\vec{\theta} \times \vec{R} \quad [7]$$

$$\vec{v} = \frac{d\vec{s}}{dt} = \frac{d\vec{\theta}}{dt} \times \vec{r} = \frac{d\vec{\theta}}{dt} \times \vec{R} \quad [8]$$

From eq. [7] and [8] we have

$$\vec{v} = \vec{\omega} \times \vec{r} = \vec{\omega} \times \vec{R} \quad [9]$$

The above equation tells us the relation between the linear and angular velocity. Now we explore relation between the linear and angular accelerations. For the purpose, differentiate the above equation with respect to time.

$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{d\vec{\omega}}{dt} \times \vec{r} + \vec{\omega} \times \frac{d\vec{r}}{dt} = \vec{\alpha} \times \vec{r} + \vec{\omega} \times \vec{v} \quad [10]$$

The first term on the RHS points along the tangent in the direction of the velocity vector and it is known as tangential acceleration $\vec{a}_T$ same as we have in circular motion. In addition, the second term point towards the center C. It is known as centripetal acceleration or normal component $\vec{a}_n$ of acceleration same as in circular motion. Now we have

$$\text{Tangential acceleration} \quad \vec{a}_T = \vec{\alpha} \times \vec{r} \quad [11]$$

$$\text{Normal acceleration} \quad \vec{a}_n = \vec{\omega} \times \vec{v} = -\omega^2 \vec{r} \quad [12]$$

2.0 MOMENT OF INERTIA (I)

Definition: Moment of inertia is defined as the capability of system to oppose the change produced in the rotational motion of a body. As “Inertia” plays an important role in definition of Newton's first law which is also called as inertia law, moment of inertia is the key concept in defining the state of rotation. It is the property of a body rotating or which can rotate about an axis and which resists the change in state of body's rotational motion.

Moment of Inertia depends on

- (i) density of the material of body
- (ii) shape & size of body
- (iii) axis of rotation

Combined we can say that it depends upon distribution of mass relative to axis of rotation.

Note : Moment of inertia does not change if the mass is rotated with constant radius about axis of rotation.

2.1 Moment of Inertia of System of Particles

Moment of inertia of system of particles is given by

$$I = mr_1^2 + m_2 r_2^2 + \dots = \sum_{i=1}^n m_i r_i^2$$

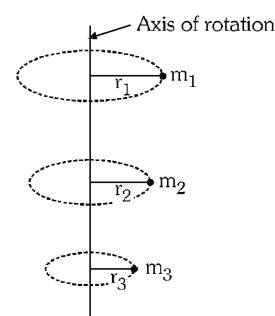
$$= I_1 + I_2 + I_3 + \dots$$

SI units of Moment of Inertia is Kgm^2 .

Moment of Inertia of a single particle : $I = mr^2$

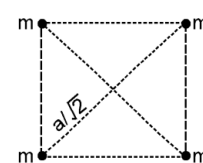
where m = mass of the particle

r = perpendicular distance of the particle from the axis about which moment of Inertia is to be calculated



Illustrations

Illustration 7. Four particles each of mass m are kept at the four corners of a square of edge a . Find the moment of inertia of the system about a line perpendicular to the plane of the square and passing through the centre of the square.



Solution The perpendicular distance of every particle from the given line is $a/\sqrt{2}$. The moment of inertia of one particle is, therefore, $m(a/\sqrt{2})^2 = \frac{1}{2}ma^2$. The moment of inertia of the system is, therefore, $4 \times \frac{1}{2}ma^2 = 2ma^2$.

Illustration 8. Two heavy particles having masses m_1 & m_2 are situated in a plane perpendicular to line AB at a distance of r_1 and r_2 respectively.

- What is the moment of inertia of the system about axis AB?
- What is the moment of inertia of the system about an axis passing through m_1 and perpendicular to the line joining m_1 and m_2 ?
- What is the moment of inertia of the system about an axis passing through m_1 and m_2 ?

Solution

- Moment of inertia of particle on left is $I_1 = m_1 r_1^2$.

Moment of Inertia of particle on right is $I_2 = m_2 r_2^2$.

Moment of Inertia of the system about AB is

$$I = I_1 + I_2 = m_1 r_1^2 + m_2 r_2^2$$

- Moment of inertia of particle on left is $I_1 = 0$

Moment of Inertia of particle on right is $I_2 = m_2(r_1 + r_2)^2$.

Moment of Inertia of the system about AB is

$$I = I_1 + I_2 = 0 + m_2(r_1 + r_2)^2$$

- Moment of inertia of particle on left is $I_1 = 0$

Moment of Inertia of particle on right is $I_2 = 0$

Moment of Inertia of the system about AB is $I = I_1 + I_2 = 0 + 0$.

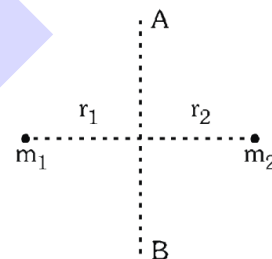
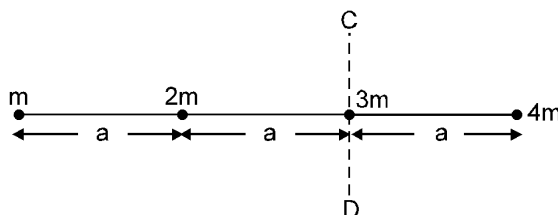
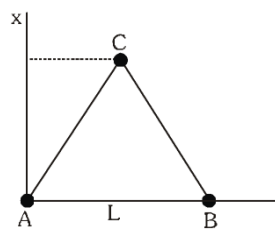


Illustration 9. Four point masses are connected by a massless rod as shown in figure. Find out the moment of inertia of the system about axis CD?



Solution $I = m(2a)^2 + 2m(a)^2 + 3m(0)^2 + 4m(a)^2 = 10ma^2$

Illustration 10. Three particles, each of mass m , are situated at the vertices of an equilateral triangle ABC of side L (figure). Find the moment of inertia of the system about the line AX perpendicular to AB in the plane of ABC .



Solution Perpendicular distance of A from $AX = 0$

Perpendicular distance of B from $AX = L$

Perpendicular distance of C from $AX = L/2$

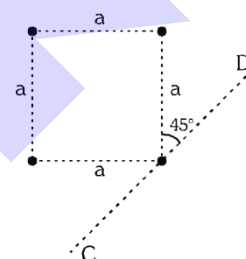
Thus, the moment of inertia of the particle at $A = 0$, of the particle $B = mL^2$, and of the particle $C = m(L/2)^2$. The moment of inertia of the three-particle system about AX is

$$I = 0 + mL^2 + m(L/2)^2 = \frac{5mL^2}{4}$$

Illustration 11. Four point masses each of mass m kept at the four corners of a square of side length 'a' find the moment of system inertia about axis CD .

Solution

$$I = 0 + m\left(\frac{a}{\sqrt{2}}\right)^2 + m\left(\frac{a}{\sqrt{2}}\right)^2 + m(\sqrt{2}a)^2$$

$$= 3ma^2$$


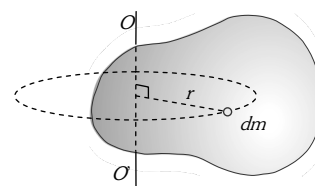
2.2 Moment of inertia of a rigid body

A rigid body is continuous distribution of mass and can be assumed consisting of infinitely large number of point particles. If one of the point particle of infinitely small mass dm is at a distance r from the axis of rotation OO' , the moment of inertia of this point particle is given by

$$dI_o = r^2 dm$$

The moment of inertia of the whole body about the axis OO' can now be obtained by integrating term of the above equation over the limits to cover whole of the body.

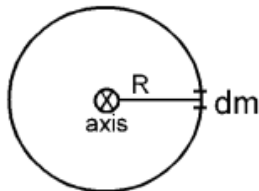
$$I_o = \int dI_o = \int r^2 dm \quad [3]$$



Expression for moment of inertia contains product of two terms. One of them is the mass of the body and the other is a characteristic dimension, which depends on the manner how mass of the body is distributed relative to the axis of rotation. Therefore moment of inertia of a rigid body depends on the mass of the body and distribution of the mass relative to the axis of rotation. Obviously for uniform bodies expression of moment of inertia depends on their shape and location and orientation of the axis of rotation. Based on these facts we can conclude.

Illustrations

Illustration 12. Calculate the moment of inertia of a ring having mass M , radius R and having uniform mass distribution about an axis passing through the centre of ring and perpendicular to the plane of ring ?



Solution

$$I = \int (dm)r^2$$

Because each element is equally distanced from the axis so $r = R = R^2 \int dm = MR^2$

$$I = MR^2$$

(**Note :** Answer will remain same even if the mass is nonuniformly distributed because $\int dm = M$ always.)

Illustration 13. Calculate the moment of inertia of a uniform rod of mass M and length ℓ about an axis passing through an end and perpendicular to the rod.

Solution

$$I = \int (dm)r^2 = \int_0^\ell \left(\frac{M}{\ell} dx \right) x^2 = \frac{M\ell^2}{3}$$

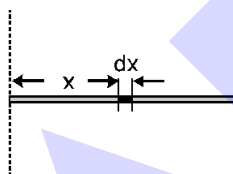


Illustration 14. Determine the moment of Inertia of a uniform disc having mass M , radius R about an axis passing through centre & perpendicular to the plane of disc ?

Solution

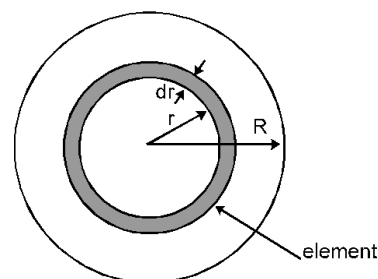
$$I = \int dI_{\text{ring}}$$

element - ring $dI = dmr^2$

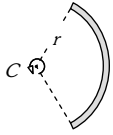
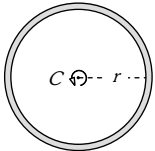
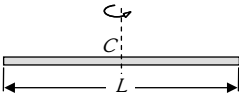
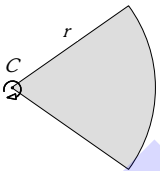
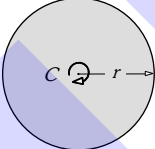
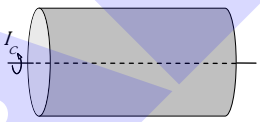
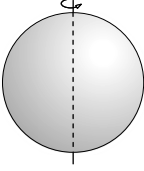
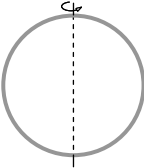
$dm = \frac{M}{\pi R^2} 2\pi r dr$ (here we have used the uniform mass distribution)

$$\therefore I = \int_0^R \frac{M}{\pi R^2} \cdot (2\pi r dr) \cdot r^2$$

$$\Rightarrow I = \frac{MR^2}{2}$$



Moment of Inertia for some commonly used bodies

Body		Axis	Moment of Inertia
Uniform thin rod bent into shape of an arc of mass m		Passing through center and perpendicular to the plane containing the arc	$I_C = mr^2$
Uniform ring of mass m		Passing through center and perpendicular to the plane containing the arc or the centroidal axis.	$I_C = mr^2$
Straight uniform rod		Passing through center and perpendicular to the rod or the centroidal axis.	$I_C = \frac{mL^2}{12}$
Sector of a uniform disc of mass m		Passing through center and perpendicular to the plane containing the sector.	$I_C = \frac{mr^2}{2}$
Uniform disc of mass m		Passing through center and perpendicular to the plane containing the disc or the centroidal axis.	$I_C = \frac{mr^2}{2}$
Homogeneous cylinder of mass m		Axis of the cylinder or the centroidal axis.	$I_C = \frac{mr^2}{2}$
Homogeneous sphere of mass m		Diameter or the centroidal axis	$I_C = \frac{2}{5}mR^2$
Spherical shell of mass m		Diameter or the centroidal axis	$I = \frac{2}{3}mR^2$

2.3 Theorems on Moment of Inertia

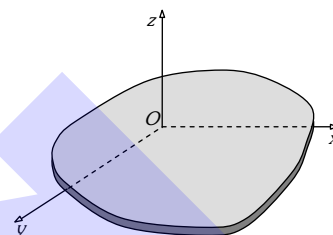
Moment of inertias of a rigid body about different axes may be different. There are two theorems known as theorem of perpendicular axes and theorem of parallel axes, which greatly simplify calculation of moment of inertia about an axis if moment of inertia of a body about another suitable axis is known.

Theorem of Perpendicular Axes

This theorem is applicable for a rigid body that lies entirely within a plane i.e. a laminar body or a rod bent into shape of a plane curve. The moment of inertia I_x , I_y and I_z of the body about the x, y and z-axis can be expressed by the following equations.

$$I_z = I_y + I_x$$

For a planar body, the moment of inertia about an axis perpendicular to the plane of the body is the sum of the moment of inertias about two perpendicular axes in the plane of the object provided that all the three axes are concurrent.

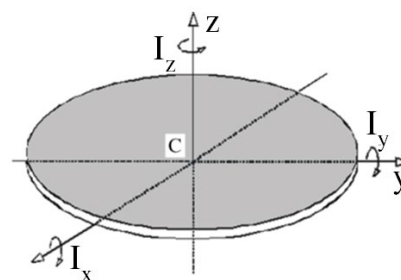


Illustrations

Illustration 15. Find moment of inertia of a uniform disc of mass m and radius r about one of its diameter.

Solution.

In the adjoining figure a disc is shown with two of its diameter perpendicular to each other. These diameters are along the x and the y-axis of a coordinate system. The x-axis is perpendicular to the plane of the disc and passes through its center is also shown.



Since the disc is symmetric about both the diameters, moment of inertias about both the diameters must be equal. Thus substituting this in the theorem of perpendicular axes, we have

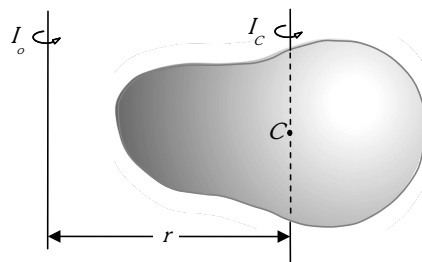
$$I_z = I_y + I_x \rightarrow I_z = 2I_x + 2I_y$$

Moment of inertia of the disc about the z-axis is $I_z = \frac{1}{2}mr^2$. Substituting it in the above equation, we have

$$I_x = I_y = \frac{1}{2}I_z = \frac{1}{4}mr^2$$

Theorem of Parallel Axes

This theorem also known as Steiner's theorem can be used to determine the moment of inertia of a rigid body about any axis, if the moment of inertia of the body about a parallel axis passing through mass center of the body and perpendicular distance between both the axes is known. Consider a body of arbitrary shape and mass m shown in the figure. Its moment of inertia I_o and I_C are defined about two parallel axes. The



axis about which moment of inertia I_C is defined passes through the mass center C . Separation between the axes is r . These two moment of inertias are related by the following equation.

$$I_o = I_C + Mr^2$$

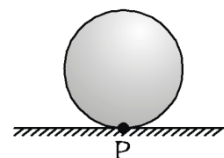
The above equation is known as the theorem of parallel axes or Steiner's theorem.

- The moment of inertia about any axis parallel to an axis through the mass center is given by sum of moment of inertia about the axis through the mass center and product term of mass of the body and square of the distance between the axes.
- Among all the parallel axes the moment of inertia of a rigid body about the axis through the mass center is the minimum moment of inertia.

The second term added to the moment of inertia I_C about the centroidal axis in the above equation can be recognized as the moment of inertia of a particle of mass equal to that of the body and located at its mass center. It again reveals that the plane motion of a rigid body is superposition of pure rotation about the mass center or centroidal rotation and translation of its mass center.

Illustrations

Illustration 16. Find moment of inertia of a uniform ring, uniform disc, uniform cylinder and uniform sphere each of mass m and radius r about an axis passing through point P and perpendicular to plane of paper.



Solution. I_C is the moment of inertia about an axis passing through centre and perpendicular to plane of paper.

Ring $I_P = I_C + Mr^2 = Mr^2 + Mr^2 = 2Mr^2$

Disc $I_P = I_C + Mr^2 = \frac{1}{2}Mr^2 + Mr^2 = \frac{3}{2}Mr^2$

Cylinder $I_P = I_C + Mr^2 = \frac{1}{2}Mr^2 + Mr^2 = \frac{3}{2}Mr^2$

Sphere $I_P = I_C + Mr^2 = \frac{2}{5}Mr^2 + Mr^2 = \frac{7}{5}Mr^2$

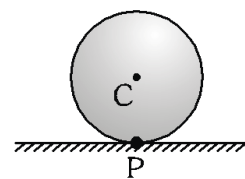
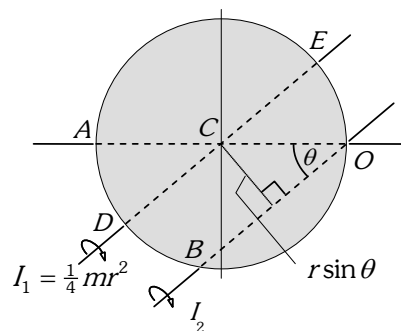


Illustration 17. Find expression for moment of inertia of a uniform disc of mass m , radius r about one of its secant making an angle θ with one of its diameter.

Solution.

A disc, the secant OB and diameter OA are shown in the adjoining figure. The secant OB is parallel to another diameter DE . Moment of inertia of the disc about one of its diameter is $\frac{1}{4}mr^2$ and hence moment of inertia I_1 about the diameter DE . Distance between the secant OB and the parallel diameter DE is $r\sin\theta$.



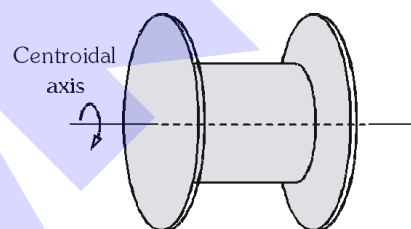
Substituting above information in the theorem of parallel axes, we have

$$I_2 = I_1 + m(r\sin\theta)^2 \rightarrow I_2 = mr^2\left(\frac{1}{4} + \sin^2\theta\right)$$

Illustration 18. Find moment of inertia about centroidal axis of a bobbin, which is constructed by joining coaxially two identical discs each of mass m and radius $2r$ to a cylinder of mass m and radius r as shown in the figure.

Solution.

The bobbin is a composite body made by joining two identical discs coaxially to cylinder. The moment of inertia I of the bobbin equals to the sum of moment of inertias of the two discs and moment of inertia of the cylinder about their centroidal axes. Using expressions for the moment of inertia for disc and cylinder, we have

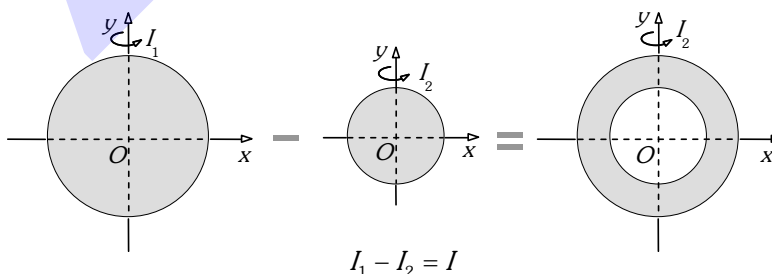


$$I = 2I_{\text{disc}} + I_{\text{cylinder}} \rightarrow I = 2\left\{\frac{1}{2}m(2r)^2\right\} + \frac{1}{2}mr^2 = \frac{9}{2}mr^2$$

Illustration 19. Find moment of inertia about one of diameter of a hollow sphere of mass m , inner radius r and outer radius R .

Solution.

The hollow sphere is assumed as if a concentric smaller sphere of radius r is removed from a larger sphere of radius R . Thus the moment inertia of the hollow sphere about any axes can be obtained by subtracting moment of inertia of the smaller sphere from that of the larger sphere. As shown in the following figure.



Let the mass of the hollow sphere is m .

Density of the material used is

$$\rho = \frac{3m}{4\pi(R^3 - r^3)}$$

Masses m_1 and m_2 of the smaller spheres are

$$m_1 = \rho \left(\frac{4}{3} \pi R^3 \right) = \frac{mR^3}{R^3 - r^3} \quad \text{and} \quad m_2 = \rho \left(\frac{4}{3} \pi r^3 \right) = \frac{mr^3}{R^3 - r^3}$$

Subtracting I_2 from I_1 we have I .

$$I = I_1 - I_2 \rightarrow I = \frac{2m_1 R^2}{5} - \frac{2m_2 r^2}{5} = \frac{2m(R^5 - r^5)}{5(R^3 - r^3)}$$

2.4 Radius of Gyration

It is the radial distance from a rotation axis at which the mass of an object could be concentrated without altering the moment of inertia of the body about that axis.

If the mass m of the body were actually concentrated at a distance k from the axis, the moment of inertia about that axis would be mk^2 .

$$k = \sqrt{\frac{I}{m}}$$

The radius of gyration has dimensions of length and is measured in appropriate units of length such as meters.

3.0 TORQUE

Torque represents the capability of a force to produce change in the rotational motion of the body.

3.1 Torque about a point

Torque of force $\vec{F}$ about a point $\vec{\tau} = \vec{r} \times \vec{F}$

Where $\vec{F}$ = force applied

P = point of application of force

Q = Point about which we want to calculate the torque.

$\vec{r}$ = position vector of the point of application of force
w.r.t. the point about which we want to determine the torque.

$$|\vec{\tau}| = r F \sin\theta = r_{\perp} F = r F_{\perp}$$

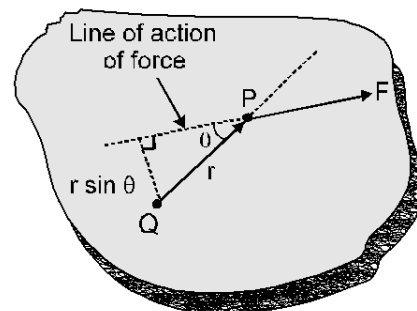
Where θ = angle between the direction of force and the position vector of P wrt. Q .

$r_{\perp} = r \sin \theta$ = perpendicular distance of line of action of force from point Q , it is also called force arm.

$F_{\perp} = F \sin \theta$ = component of $\vec{F}$ perpendicular to $\vec{r}$

SI unit of torque is N-m

Torque is a vector quantity and its direction is determined using right hand thumb rule and its always perpendicular to the plane of rotation of the body.



Illustrations

Illustration 20. A particle of mass M is released in vertical plane from a point P at $x = x_0$ on the x -axis it falls vertically along the y -axis. Find the torque τ acting on the particle at a time t about origin ?

Solution Torque is produced by the force of gravity.

$$\vec{\tau} = r F \sin \theta \hat{k}$$

$$\text{or } \tau = r_{\perp} F = x_0 mg$$

$$\vec{\tau} = mgx_0 \hat{k}$$

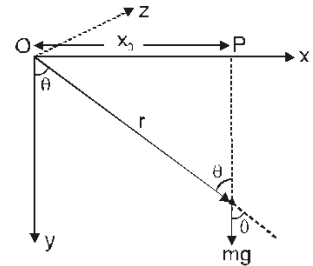


Illustration 21. A particle having mass m is projected with a velocity v_0 from a point P on a horizontal ground making an angle θ with horizontal. Find out the magnitude of torque about the point of projection acting on the particle when it is at its maximum height ?

Solution $\tau = rF \sin \theta = \frac{R}{2} mg = \frac{v_0^2 \sin 2\theta}{2g} mg \Rightarrow \tau = \frac{mv_0^2 \sin 2\theta}{2}$

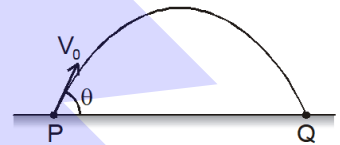
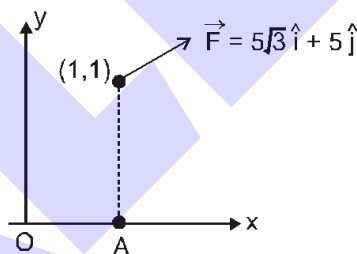


Illustration 22. Find the torque about point O and A due to given force $\vec{F}$.



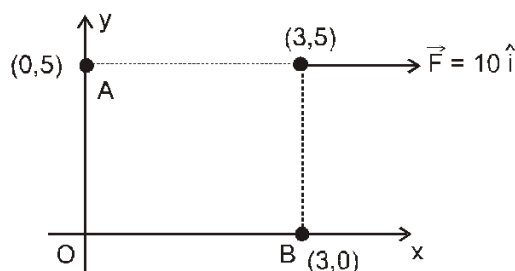
Solution Torque about point O , $\vec{\tau} = \vec{r}_0 \times \vec{F}$, $\vec{r}_0 = \hat{i} + \hat{j}$, $\vec{F} = 5\sqrt{3}\hat{i} + 5\hat{j}$

$$\vec{\tau} = (\hat{i} + \hat{j}) \times (5\sqrt{3}\hat{i} + 5\hat{j}) = 5(1 - \sqrt{3})\hat{k}$$

Torque about point A , $\vec{\tau} = \vec{r}_a \times \vec{F}$, $\vec{r}_a = \hat{j}$, $\vec{F} = 5\sqrt{3}\hat{i} + 5\hat{j}$

$$\vec{\tau} = \hat{j} \times (5\sqrt{3}\hat{i} + 5\hat{j}) = 5(-\sqrt{3})\hat{k}$$

Illustration 23. Find out torque about point A , O and B due to given force $\vec{F}$.



Solution Torque about point A, $\vec{\tau}_A = \vec{r}_A \times \vec{F}$, $\vec{r}_A = 3\hat{i}$, $\vec{F} = 10\hat{i}$
 $\vec{\tau}_A = 3\hat{i} \times 10\hat{i} = 0$
 Torque about point B, $\vec{\tau}_B = \vec{r}_B \times \vec{F}$, $\vec{r}_B = 5\hat{j}$, $\vec{F} = 10\hat{i}$
 $\vec{\tau}_B = 5\hat{j} \times 10\hat{i} = -50\hat{k}$
 Torque about point O, $\vec{\tau}_O = \vec{r}_O \times \vec{F}$, $\vec{r}_O = 3\hat{i} + 5\hat{j}$, $\vec{F} = 10\hat{i}$
 $\vec{\tau}_O = (3\hat{i} + 5\hat{j}) \times 10\hat{i} = -50\hat{k}$

3.2 Torque about an axis

The torque of a force $\vec{F}$ about an axis AB is defined as the component of torque of $\vec{F}$ about any point O on the axis AB, along the axis AB.

In the given figure torque of $\vec{F}$ about O is $\vec{\tau}_O = \vec{r} \times \vec{F}$

The torque of $\vec{F}$ about AB, τ_{AB} is component of $\vec{\tau}_O$ along line AB.

There are four cases of torque of a force about an axis.:

Case I : Force is parallel to the axis of rotation, $\vec{F} \parallel \overline{AB}$. AB is the axis of rotation about which torque is required $\vec{r} \times \vec{F}$ is perpendicular to $\vec{F}$, but $\vec{F} \parallel \overline{AB}$, hence $\vec{r} \times \vec{F}$ is perpendicular to $\overline{AB}$. The component of $\vec{r} \times \vec{F}$ along $\overline{AB}$ is, therefore, zero.

Case II : The line of force intersects the axis of rotation (F intersect AB)

$\vec{F}$ intersects AB, then $\vec{F}$ and $\vec{r}$ are along the same line.

The torque about O is

$$\vec{r} \times \vec{F} = 0.$$

Hence component this torque along line AB is also zero.

Case III : $\vec{F}$ perpendicular to $\overline{AB}$ but $\vec{F}$ and AB do not intersect.

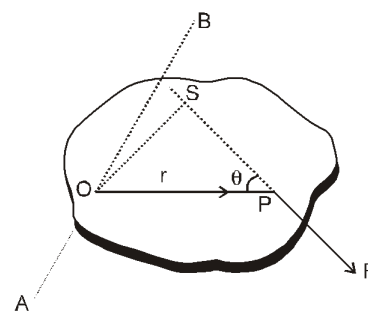
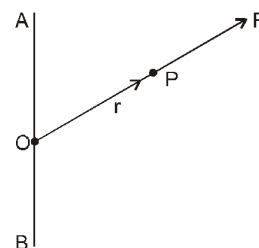
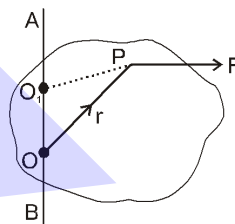
In the three dimensions, two lines may be perpendicular without intersecting each other.

Two nonparallel and nonintersecting lines are called skew lines.

Figure shows the plane through the point of application of force P that is perpendicular to the axis of rotation AB. Suppose the plane intersects the axis at the point O. The force F is in this plane (since F is perpendicular to AB). Taking the origin at O,

$$\text{Torque} = \vec{r} \times \vec{F} = \overline{OP} \times \vec{F}.$$

$$\text{Thus, torque} = rF \sin \theta = F(OS)$$



where OS is the perpendicular from O to the line of action of the force $\vec{F}$. The line OS is also perpendicular to the axis of rotation. It is thus the length of the common perpendicular to the force and the axis of rotation.

The direction of $\vec{\tau} = \vec{OP} \times \vec{F}$ is along the axis AB because $\vec{AB} \perp \vec{OP}$ and $\vec{AB} \perp \vec{F}$. The torque about AB is, therefore, equal to the magnitude of $\vec{\tau}$ that is $F \cdot (OS)$.

Thus, the torque of F about AB = magnitude of the force F $\times$ length of the common perpendicular to the force and the axis. The common perpendicular OS is called the lever arm or moment arm of this torque.

Case IV : $\vec{F}$ and $\vec{AB}$ are skew but not perpendicular.

Here we resolve $\vec{F}$ into two components, one is parallel to axis and other is perpendicular to axis. Torque of the parallel part is zero and that of the perpendicular part may be found, by using the result of **case (III)**.

Illustrations

Illustration 24. Find the torque of weight about the axis passing through point P.

Solution

$$\vec{\tau} = \vec{r} \times \vec{F}, \quad \vec{r} = R$$

r and $mg \sin \theta$ both are at perpendicular so torque about point

$$P = mg R \sin \theta$$

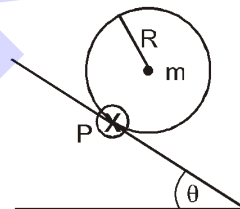


Illustration 25. A bob of mass m is suspended at point O by string of length ℓ . Bob is moving in a horizontal circle find out (i) magnitude of torque of gravity and tension about point O and O'. (ii) Magnitude of net torque about axis OO'.

Solution

(i) Torque about point O

$$\text{Torque of tension (T), } \tau_{\text{ten}} = 0$$

(tension is passing through point O)

$$\text{Torque of gravity } \tau_{\text{mg}} = \ell mg \sin \theta$$

Torque about point O'

$$\text{Torque of gravity } \tau_{\text{mg}} = mgr, \quad r = \ell \sin \theta$$

Torque of tension

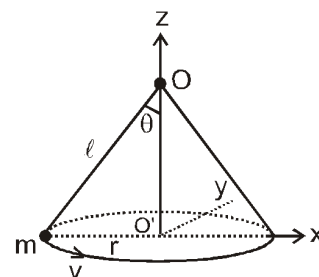
$$\tau_{\text{ten}} = \frac{mg}{\cos \theta} (\ell \sin \theta) \cos \theta = mg \ell \sin \theta \text{ (along positive } \hat{j} \text{)}$$

(ii) Torque about axis OO'

$$\text{Torque of gravity about axis OO'} \tau_{\text{mg}} = 0 \text{ (force mg parallel to axis OO')}$$

$$\text{Torque of tension about axis OO'} \tau_{\text{ten}} = 0 \text{ (force T is passing through the axis OO')}$$

$$\text{Net torque about axis OO'} \tau_{\text{net}} = 0$$

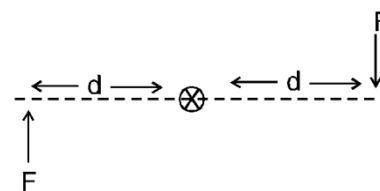


3.3 Force Couple

A pair of forces each of same magnitude and acting in opposite direction is called a force couple.

Torque due to couple = Magnitude of one force $\times$ distance between their lines of action.

Magnitude of torque = $\tau = F(2d)$

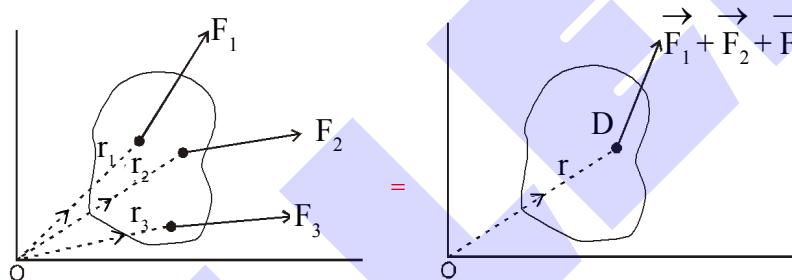


- A couple does not exert a net force on an object even though it exerts a torque.
- Net torque due to a force couple is same about any point.
- A consequence is that, if $F_{\text{net}} = 0$ and $\tau_{\text{net}} = 0$ about one point, then $\tau_{\text{net}} = 0$ about any point.

3.4 Point of Application of Force

Point of Application of force is the point at which, if net force is assumed to be acting, then it will produce same translational as well as rotational effect, as was produced earlier.

We can also define point of application of force as a point about which torque of all the forces is zero.

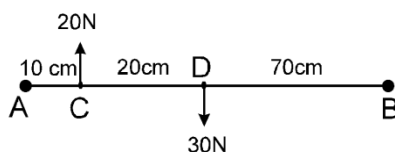


Consider three forces $\vec{F}_1, \vec{F}_2, \vec{F}_3$ acting on a body if D is point of application of force then torque of $\vec{F}_1 + \vec{F}_2 + \vec{F}_3$ acting at a point D about O is same as the original torque about O

$$[\vec{r}_1 \times \vec{F}_1 + \vec{r}_2 \times \vec{F}_2 + \vec{r}_3 \times \vec{F}_3] = \vec{r} \times (\vec{F}_1 + \vec{F}_2 + \vec{F}_3)$$

Illustrations

Illustration 26. Determine the point of application of force, when forces of 20 N & 30 N are acting on the rod as shown in figure.



Solution Net force acting on the rod $F_{\text{net}} = 10\text{ N}$

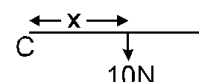
Net torque acting on the rod about point C

$$\tau_c = (20 \times 0) + (30 \times 20) = 600 \text{ clockwise}$$

Let the point of application be at a distance x from point C

$$600 = 10x \Rightarrow x = 60 \text{ cm}$$

$\therefore$ 70 cm from A is point of Application



Note :

- (i) Point of application of gravitational force is known as the centre of gravity.
- (ii) Centre of gravity coincides with the centre of mass if value of $\bar{g}$ is assumed to be constant.
- (iii) Concept of point of application of force is imaginary, as in some cases it can lie outside the body.

3.5 Equilibrium

A system is in mechanical equilibrium if it is in translational as well as rotational equilibrium.

For this : $\vec{F}_{\text{net}} = 0$

$\vec{\tau}_{\text{net}} = 0$ (about every point)

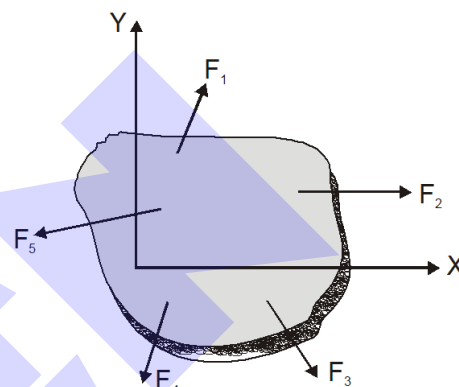
if $\vec{F}_{\text{net}} = 0$ then $\vec{\tau}_{\text{net}}$ is same about every point

Hence necessary and sufficient condition for equilibrium is

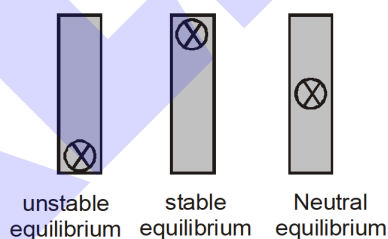
$\vec{F}_{\text{net}} = 0$,

$$\Rightarrow \vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \vec{F}_4 + \vec{F}_5 = 0$$

$\vec{\tau}_{\text{net}} = 0$ about any one point, which we can choose as per our convenience. ($\vec{\tau}_{\text{net}}$ will automatically be zero about every point)

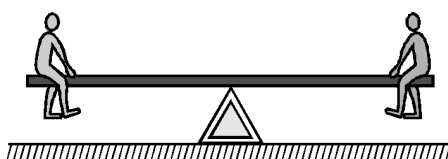


- The equilibrium of a body is called **stable** if the body tries to regain its equilibrium position after being slightly displaced and released. It is called **unstable** if it gets further displaced after being slightly displaced and released. If it can stay in equilibrium even after being slightly displaced and released, it is said to be in **neutral** equilibrium.



Illustrations

Illustration 27. Two small kids weighing 10 kg and 15 kg are trying to balance a seesaw of total length 5.0 m, with the fulcrum at the centre. If one of the kids is sitting at an end, where should the other sit ?



Solution It is clear that the 10 kg kid should sit at the end and the 15 kg kid should sit closer to the centre. Suppose his distance from the centre is x . As the kids are in equilibrium, the normal force between a kid and the seesaw equals the weight of that kid. Considering the rotational equilibrium of the seesaw, the torque of the forces acting on it should add to zero. The forces are

- (a) $(15 \text{ kg})g$ downward by the 15 kg kid,
- (b) $(10 \text{ kg})g$ downward by the 10 kg kid,
- (c) weight of the seesaw and
- (d) the normal force by the fulcrum.

Taking torques about the fulcrum,

$$(15 \text{ kg})g x = (10 \text{ kg})g (2.5 \text{ m}) \quad \text{or} \quad x = 1.7 \text{ m}.$$

Illustration 28. A uniform ladder of mass $m = 10 \text{ kg}$ leans against a smooth vertical wall making an angle $\theta = 53^\circ$ with it. The other ends rests on a rough horizontal floor. Find the normal force and the friction force that the floor exerts on the ladder.

Solution The forces acting on the ladder are shown in figure. They are

- (a) Its weight W ,
- (b) normal force N_1 by the vertical wall,
- (c) normal force N_2 by the floor and
- (d) frictional force f by the floor.

Taking horizontal and vertical components,

$$N_1 = f \quad \dots(i)$$

$$\text{and } N_2 = mg \quad \dots(ii)$$

Taking torque about B,

$$N_1(AO) = mg(CB)$$

$$\text{or, } N_1(AB) \cos \theta = mg \frac{AB}{2} \sin \theta$$

$$\text{or } N_1 \frac{3}{5} = \frac{W}{2} \frac{4}{5}$$

$$\text{or, } N_1 = \frac{2}{3}W \quad \dots(iii)$$

The normal force by the floor is $N_2 = W = (10 \text{ kg}) (9.8 \text{ m/s}^2) = 98 \text{ N}$.

The frictional force is $f = N_1 = \frac{2}{3}W = 65 \text{ N}$.

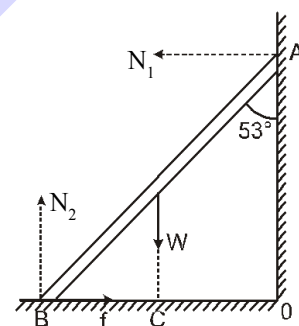
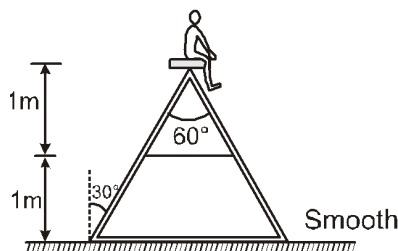


Illustration 29. The ladder shown in figure has negligible mass and rests on a frictionless floor. A crossbar connects the two legs of the ladder at the centre as shown. The angle between the two legs is 60° . The fat person sitting on the ladder has a mass of 80 kg . Find the constant forces exerted by the floor on each leg and the tension in the crossbar.



Solution

The forces acting on different parts are shown in figure. Consider the vertical equilibrium of “the ladder plus the person” system. The forces acting on this system are its weight $(80 \text{ kg})g$ and the contact force N $+ N = 2N$ due to the floor. Thus

$$2N = (80 \text{ kg})g$$

$$\text{or } N = (40 \text{ kg})(9.8 \text{ m/s}^2) = 392 \text{ N.}$$

Next consider the equilibrium of the left leg of the ladder. Taking torques of the forces acting on it about the upper end,

$$N(2\text{m}) \tan 30^\circ = T(1 \text{ m}) \text{ or } T = N \frac{2}{\sqrt{3}} = (392 \text{ N}) \times \frac{2}{\sqrt{3}} = 450 \text{ N.}$$

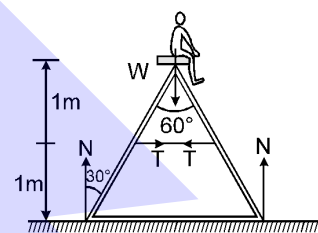


Illustration 30. A uniform rod of length ℓ , mass m is hung from two strings of equal length from a ceiling as shown in figure. Determine the tensions in the strings?

Solution

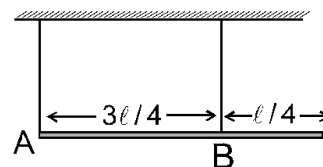
$$T_A + T_B = mg \quad \dots\dots\dots(i)$$

Torque about point A is zero

$$\text{So, } T_B \times \frac{3\ell}{4} = mg \frac{\ell}{2} \quad \dots\dots\dots(ii)$$

From eq. (i) & (ii), d

$$T_A = mg/3, \quad T_B = 2mg/3.$$



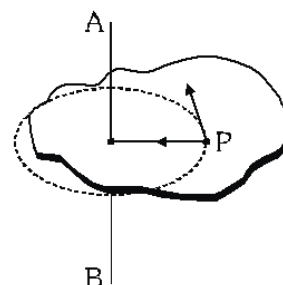
4.0 ROTATION ABOUT A FIXED AXIS

Consider a rigid body rotating about a fixed axis AB (figure). Consider a particle P of mass m rotating in a circle of radius r .

The radial acceleration of the particle $= \frac{v^2}{r} = \omega^2 r$.

(Where v is speed of particle, ω is angular speed of body at r is distance of particle from axis of rotation)

Thus, the radial force on it $= m\omega^2 r$



The tangential acceleration of the particle = $\frac{dv}{dt}$

Thus, the tangential force on it = $m \frac{dv}{dt} = mr \frac{d\omega}{dt} = mr\alpha$

The torque of $m\omega^2 r$ about AB is zero as it intersects the axis and that of $mr\alpha$ is $mr^2\alpha$ as the force and the axis are skew and perpendicular. Thus, the torque of the resultant force acting on P is $mr^2\alpha$. Summing over all the particles, the total torque of all the forces acting on all the particles of the body is

$$\tau_{\text{total}} = \sum_i m_i r_i^2 \alpha = I\alpha \quad \dots(i)$$

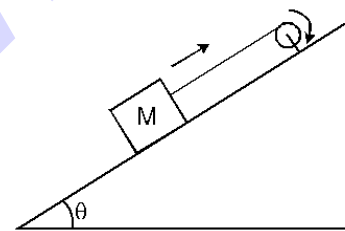
where

$$I = \sum_i m_i r_i^2$$

The quantity I is called the moment of inertia of the body about the axis of rotation. Note that m_i is the mass of the i^{th} particle and r_i is its perpendicular distance from the axis.

Illustrations

Illustration 31. A wheel of radius r and moment of inertia I about its axis is fixed at the top of an inclined plane of inclination θ as shown in figure. A string is wrapped round the wheel and its free end supports a block of mass M which can slide on the plane. Initially, the wheel is rotating at a speed ω in a direction such that the block slides up the plane. How far will the block move before stopping?



Solution

Suppose the deceleration of the block is a . The linear deceleration of the rim of the wheel is also a . The angular deceleration of the wheel is $\alpha = a/r$. If the tension in the string is T , the equations of motion are as follows:

$$Mg \sin \theta - T = Ma \quad \text{and} \quad Tr = I\alpha = Ia/r.$$

Eliminating T from these equations,

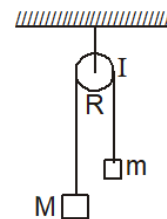
$$Mg \sin \theta - I \frac{a}{r^2} = Ma$$

$$\text{giving, } a = \frac{Mg r^2 \sin \theta}{I + Mr^2}$$

The initial velocity of the block up the incline is $v = \omega r$. Thus, the distance moved by the block before stopping is

$$x = \frac{v^2}{2a} = \frac{\omega^2 r^2 (I + Mr^2)}{2M r^2 \sin \theta} = \frac{(I + Mr^2)\omega^2}{2M g \sin \theta}$$

Illustration 32. The pulley shown in figure has a moment of inertia I about its axis and its radius is R . Find the magnitude of the acceleration of the two blocks. Assume that the string is light and does not slip on the pulley.



Solution Suppose the tension in the left string is T_1 and that in the right string is T_2 . Suppose the block of mass M goes down with acceleration a and the other block moves up with the same acceleration.

This is also the tangential acceleration of the rim of the wheel as the string does not slip over the rim. The angular acceleration of the wheel is, therefore, $\alpha = a/R$. The equations of motion for the mass M , the mass m and the pulley are as follows:

$$Mg - T_1 = Ma \quad \text{.....(i)}$$

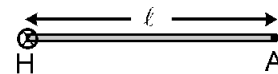
$$T_2 - mg = ma \quad \text{.....(ii)}$$

$$T_1 R - T_2 R = I\alpha = I a / R \quad \text{.....(iii)}$$

Putting T_1 and T_2 from (i) and (ii) into (iii),

$$[(Mg - a) - m(g + a)] R = I \frac{a}{R} \text{ which gives } a = \frac{(M - m)gR^2}{I + (M + m)R^2}.$$

Illustration 33. A uniform rod of mass m and length ℓ can rotate in vertical plane about a smooth horizontal axis hinged at point H.

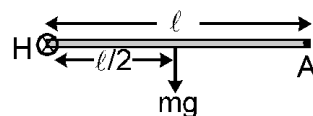


- Find angular acceleration α of the rod just after it is released from initial horizontal position from rest?
- Calculate the acceleration (tangential and radial) of point A at this moment.
- Calculate net hinge force acting at this moment.

Solution

$$(i) \tau_H = I_H \alpha$$

$$mg \cdot \frac{\ell}{2} = \frac{m\ell^2}{3} \alpha \Rightarrow \alpha = \frac{3g}{2\ell}$$



$$(ii) a_{tA} = \alpha \ell = \frac{3g}{2\ell} \cdot \ell = \frac{3g}{2}$$

$$a_{cA} = \omega^2 r = 0 \quad (\because \omega = 0 \text{ just after release})$$

- Suppose hinge exerts normal reaction in component form as shown



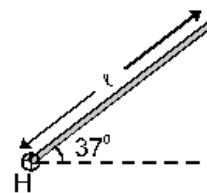
In vertical direction $F_{\text{ext}} = ma_{\text{CM}}$

$$\Rightarrow mg - N_1 = m \cdot \frac{3g}{4} \Rightarrow a_{\text{cm}} = \frac{\ell}{2} \alpha \Rightarrow N_1 = \frac{mg}{4}$$

In horizontal direction

$$F_{\text{ext}} = ma_{\text{CM}} \Rightarrow N_2 = 0 \quad (\because a_{\text{CM}} \text{ in horizontal} = 0 \text{ as } \omega = 0 \text{ just after release}).$$

Illustration 34. A uniform rod of mass m and length ℓ can rotate in vertical plane about a smooth horizontal axis hinged at point H. Find angular acceleration α of the rod just after it is released from initial position making an angle of 37° with horizontal from rest? Find force exerted by the hinge just after the rod is released from rest.



Solution Torque about hinge $= \tau_H = I \alpha$

$$mg \cos 37^\circ \frac{\ell}{2} = \frac{m\ell^2}{3} \alpha$$

$$\alpha = 6g / 5\ell$$

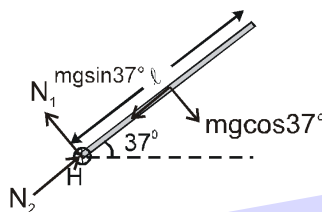
$$a_t = \alpha \frac{\ell}{2} = \frac{3g}{5}$$

$$mg \cos 37^\circ - N_1 = ma_t$$

$$N_1 = \frac{mg}{5}$$

$$\text{angular velocity of rod is zero. so } N_2 = mg \sin 37^\circ = 3mg/5$$

$$N = \sqrt{N_1^2 + N_2^2} = \sqrt{\left(\frac{mg}{5}\right)^2 + \left(\frac{3mg}{5}\right)^2} = \frac{mg\sqrt{10}}{5}$$



5.0 COMBINED TRANSLATIONAL AND ROTATIONAL MOTION OF A RIGID BODY

The general motion of a rigid body can be thought of as a sum of two independent motions. A translation of some point of the body plus a rotation about this point. A most convenient choice of the point is the centre of mass of the body as it greatly simplifies the calculations.

Consider a fan inside a train, and an observer A on the platform.

If the fan is switched off while the train moves, the motion of fan is pure translation as each point on the fan undergoes same translation in any time interval.

If fan is switched on while the train is at rest the motion of fan is pure rotation about axle ; as each point on the axle is at rest, while other points revolve about it with equal angular velocity.

if the fan is switched on while the train is moving, the motion of fan to the observer on the platform is neither pure translation nor pure rotation. This motion is an Illustration of general motion of a rigid body.

Now if there is an observer B inside the train, the motion of fan will appear to him as pure rotation.

Hence we can see that the general motion of fan w.r.t. observer A can be resolved into pure rotation of fan as observed by observer B plus pure translation of observer B (w.r.t. observer A)

Such a resolution of general motion of a rigid body into pure rotation & pure translation is not restricted to just the fan inside the train, but is possible for motion of any rigid system.

5.1 Instantaneous Axis of Rotation (IAR)

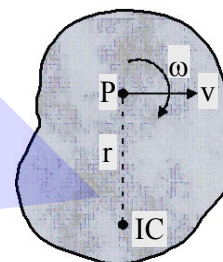
It is a mathematical line about that a body in combined translation and rotation can be conceived in pure rotation at an instant. It continuously changes its location. This axis is always perpendicular to the plane used to represent the motion and the intersection of the axis with this plane defines the location of instantaneous centre of zero velocity (IC).

Location of the Instantaneous Centre of Rotation

If the location of the IC is unknown, it may be determined by using the fact that the relative position vector extending from the IC to a point is always perpendicular to the velocity of the point. Following three possibilities exist.

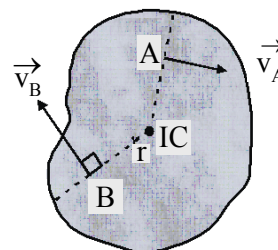
- (i) **Given the velocity of a point (normally the centre of mass) on the body and the angular velocity of the body.**

If $\vec{v}$ and ω are known, the IC is located along the line drawn perpendicular to $\vec{v}$ at P, such that the distance from P to IC is, $r = \frac{v}{\omega}$. Note that IC lie on that side of P which causes rotation about the IC, which is consistent with the direction of motion caused by $\vec{\omega}$ and $\vec{v}$



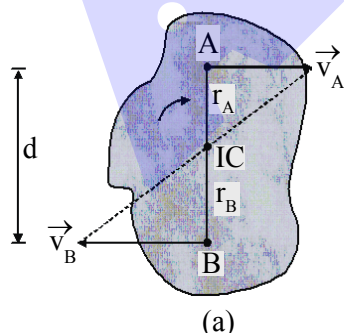
- (ii) **Given the lines of action of two non-parallel velocities**

Consider the body shown in figure where the line of action of the velocities $\vec{v}_A$ and $\vec{v}_B$ are known. Draw perpendiculars at A and B to these lines of action. The point of intersection of these perpendiculars as shown locates the IC at the instant considered.

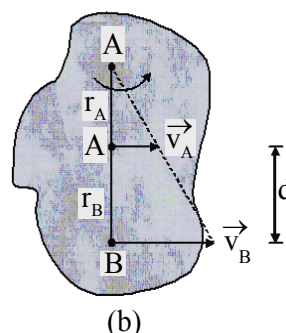


- (iii) **Given the magnitude and direction of two parallel velocities**

When the velocities of points A and B are parallel and have known magnitudes v_A and v_B then the location of the IC is determined by proportional triangles as shown in figure.



(a)



(b)

In both the cases,

$$r_A = \frac{v_A}{\omega} \text{ and } r_B = \frac{v_B}{\omega}$$

In fig. (a)

$$r_A + r_B = d$$

and in fig. (b)

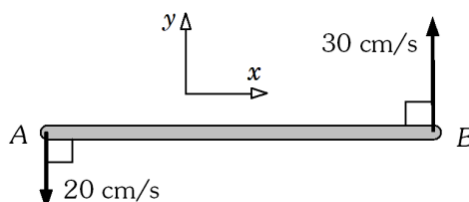
$$r_B - r_A = d$$

As a special case, if the body is translating, $v_A = v_B$ and the IC would be located at infinity, in which case $\omega = 0$.

Illustrations

Illustration 35. A 100 cm rod is moving on a horizontal surface. At an instant, when it is parallel to the x -axis its ends A and B have velocities 30 cm/s and 20 cm/s as shown in the figure.

- (a) Find its angular velocity and velocity of its center.
 (b) Locate its instantaneous axis of rotation.



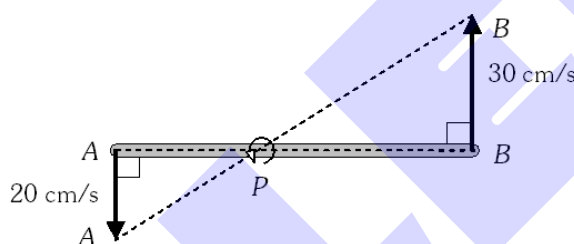
Solution.

Velocity vector of point B

$$\vec{v}_B = \vec{v}_A + \vec{\omega} \times \vec{AB} \rightarrow \omega = 0.5 \text{ rad/s}$$

Velocity vector of the center C of the rod.

$$\vec{v}_C = \vec{v}_A + \vec{\omega} \times \vec{AC} \rightarrow \vec{v}_C = -20\hat{j} + 0.5\hat{k} \times 50\hat{i} = 5.0\hat{j} \text{ cm/s}$$



(b) Here velocity vectors of the particles A and B are antiparallel, therefore the instantaneous axis of rotation passes through intersection of the common perpendicular to their velocity vectors and a line joining tips of the velocity vectors. The required geometrical construction is shown in the following figure.

Since triangles $AA'P$ and $BB'P$ are similar and $AB = 100$ cm, we $AP = 40$ have cm.

The instantaneous axis of rotation passes through the point P , which is 40 cm from A .

Analytical Approach.

The instantaneous center of rotation is at instantaneous rest.

$$\vec{v}_P = \vec{v}_A + \vec{\omega} \times \vec{AP} \rightarrow \vec{0} = -20\hat{j} + 0.5\hat{k} \times (AP)\hat{j} \Rightarrow AP = 40 \text{ cm}$$

Illustration 36. Can you suggest a quick way to find angular velocity of a rod, if velocities of two of its points are known?

Solution.

When distance between two points and their velocity components perpendicular to the lining joining them are known.

Angular velocity of the rod

$$\omega = \frac{|\vec{v}_{BA}|}{AB} = \frac{|\vec{v}_{B\perp} - \vec{v}_{A\perp}|}{AB}$$

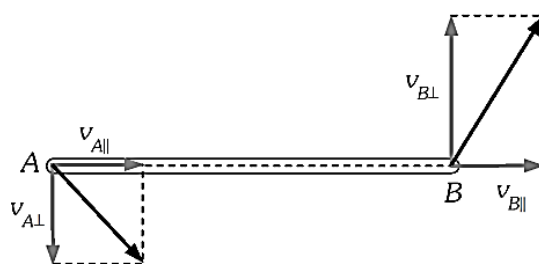
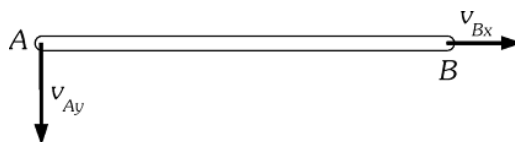


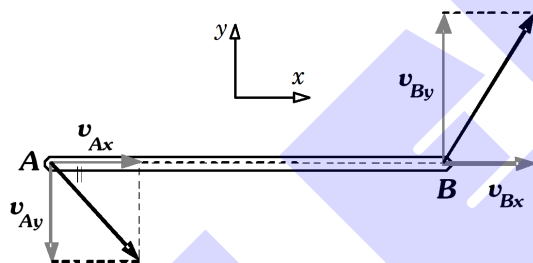
Illustration 37. A 50 cm long rod AB is in combined translation and rotation motion on a table. At an instant velocity component of point A perpendicular the rod is 10 cm/s, velocity component of point B parallel to the rod is 6.0 cm/s and angular velocity of the rod is 0.4 rad/s in anticlockwise sense as shown in the given figure.



- (a) Find velocity vectors of point A and B .
 (b) Locate the instantaneous axis of rotation.

Solution.

Let x - y plane of a coordinate system coincides with the table top and the rod is parallel to the x -axis at the instant considered. The rod is shown in this coordinate frame in the following figure.



- (a) Since distance between any two points remains unchanged, the velocity components of any two points parallel to the line joining them must be equal. Therefore, we have

$$v_{Ax} = v_{Bx} = 6.0 \text{ cm/s} \quad (1)$$

Velocities of points A and B must satisfy following equation

$$\vec{v}_B = \vec{v}_A + \vec{\omega} \times \vec{AB} \rightarrow 6.0\hat{i} + v_{By}\hat{j} = 6.0\hat{i} - 10\hat{j} + 0.4\hat{k} \times 50\hat{i}$$

Equating y -components of both the sides, we have

$$v_{By} = 10 \text{ cm/s} \quad (2)$$

From eq. (1), (2) and the given information, we can express the velocity vectors of the points A and B .

$$\vec{v}_A = 6.0\hat{i} - 10\hat{j} \text{ cm/s, and } \vec{v}_B = 6.0\hat{i} + 10\hat{j} \text{ cm/s}$$

- (b) The instantaneous axis of rotation is at instantaneous rest. Let the end A of the rod is at the origin and coordinates of the point P in the x - y plane through which the IAR passes is (x, y) .

$$\vec{v}_P = \vec{v}_A + \vec{\omega} \times \vec{AP} \rightarrow \vec{0} = 6.0\hat{i} - 10\hat{j} + 0.4\hat{k} \times (x\hat{i} + y\hat{j})$$

Equating coefficients of x and y -components of both the sides, we have

$$x = 25 \text{ cm and } y = 15 \text{ cm}$$

Therefore, coordinates of the point P through which IAR passes the x - y plane are (25, 15).

5.2 Rolling as Rotation about an Axis in Translation

Wheels of a moving vehicle roll on road. A ball rolls on ground when pushed. In fact, a body of round section can roll smoothly under favourable conditions. On the other hand, objects with corners, such as dice, roll by successive rotations about the edge or corner that is in contact with the ground. This type of motion is usually known as toppling.

If the point of contact of the rolling body does not slide it is known as *rolling without slipping* or *pure rolling* or simply *rolling* and if the point of contact slides it is known as *rolling with slipping*.

All kind of rolling motion is examples of rotation about an axis in translation.

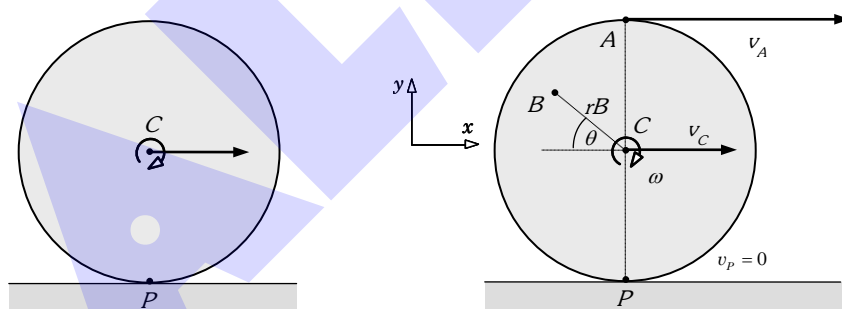
Rolling without slipping on stationary surface.

We first discuss velocity relations and thereafter accelerations relations of two points of a body of round section rolling on a stationary surface. For the purpose, we can use any of the following methods.

- I Analytical Method: By using relative motion equations.
- II Superposition Method: By superimposing translation of a point and pure rotation about that point.
- III Use of ICR.

Velocity relations by Analytical Method

Its point of contact P does not slide on the surface, therefore velocity of the point of contact relative to the surface is zero. In the next figure, velocity vectors of its center C and top point A are shown.



Velocity of the center C can be obtained with the help of relative motion equation.

$$\vec{v}_C = \vec{v}_P + \vec{\omega} \times \vec{PC} \rightarrow \vec{v}_C = \vec{0} + (-\omega \hat{k}) \times R \hat{j}$$

$$\vec{v}_C = \omega R \hat{i}$$

The above equation is used as condition of rolling without slipping on stationary surface.

Velocity of the top point A can be obtained by relative motion equation.

$$\vec{v}_A = \vec{v}_P + \vec{\omega} \times \vec{PA} \rightarrow \vec{v}_A = \vec{0} + (-\omega \hat{k}) \times (2R \hat{j})$$

$$\vec{v}_A = 2\omega R \hat{i} = 2\vec{v}_C$$

Once velocity of the center is obtained, we can use relative motion between A and C as well.

$$\vec{v}_A = \vec{v}_C + \vec{\omega} \times \vec{CA} \rightarrow \vec{v}_C = \omega R \hat{i} + (-\omega \hat{k}) \times (R \hat{j})$$

$$\vec{v}_A = 2\omega R \hat{i} = 2\vec{v}_C$$

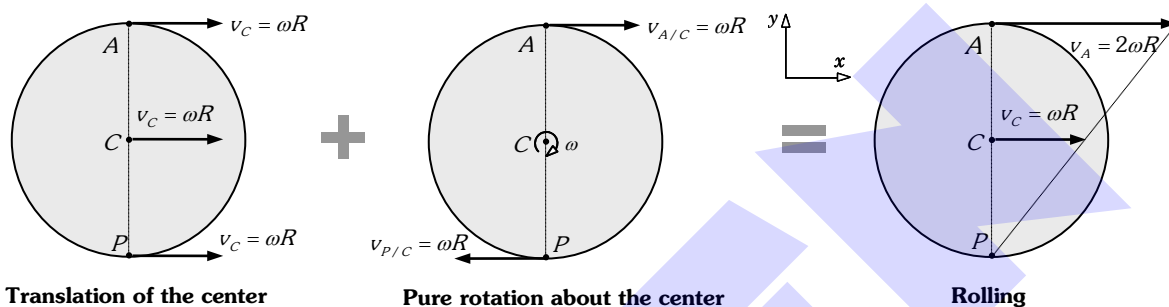
In similar fashion, velocity vector of an arbitrarily chosen point B .

$$\vec{v}_B = \vec{v}_C + \vec{\omega} \times \vec{CB} \rightarrow \vec{v}_B = v_C \hat{i} + (-\omega \hat{k}) \times (-r \cos \theta \hat{i} + r \sin \theta \hat{j})$$

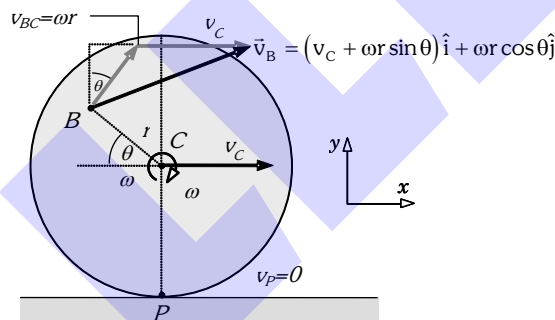
$$\vec{v}_B = (v_C + \omega r \sin \theta) \hat{i} + \omega r \cos \theta \hat{j}$$

Velocity relations by Superposition Method

Now we will see that the above velocity relation can also be obtained by assuming rolling of the wheel as superposition of translation of its center and simultaneous rotation about the center.

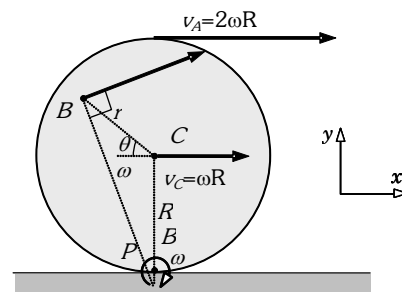


Velocity of an arbitrary point B as superposition of translation of the center and rotation about the center.



Velocity relations by Use of ICR

In rolling without slipping on stationary surface the point of contact is at instantaneous rest, therefore the ICR and the IAR passes through it. We will see how velocity of the center C , the top point A and an arbitrarily chosen point B can be calculated by assuming the body in state of pure rotation about the ICR.



Velocity of center C

$$\vec{v}_C = \vec{\omega} \times \vec{PC} = \omega R \hat{i}$$

Velocity of the top point A

$$\vec{v}_A = \vec{\omega} \times \vec{PA} = 2\omega R \hat{i}$$

Velocity of the point B

$$\vec{v}_B = \vec{\omega} \times \vec{PB} = (v_C + \omega r \sin \theta) \hat{i} + \omega r \cos \theta \hat{j}$$

Illustrations

Illustration 38. A cylinder of radius 5 m rolls on a horizontal surface. Velocity of its center is 25 m/s.

Find its angular velocity and velocity of the point A .

Solution

In rolling the angular velocity $\vec{\omega}$ and velocity of the center of a round section body satisfy condition

$$\vec{v}_C = \vec{\omega} \times \vec{r}_{C/P} \rightarrow 25\hat{i} = \omega\hat{k} \times 5\hat{j} \Rightarrow \vec{\omega} = -5\hat{k} \text{ rad/s}$$

Angular velocity vector points in the negative z -axis so the cylinder rotates in clockwise sense.

Velocity of the point A can be calculated by either analytical method, superposition method or by using method of ICR.

Analytical Method

$$\vec{v}_A = \vec{v}_C + \vec{\omega} \times \vec{CA} \rightarrow \vec{v}_A = 25\hat{i} + (-5\hat{k}) \times (-5\cos 37^\circ\hat{i} + 5\sin 37^\circ\hat{j})$$

$$\vec{v}_A = (40\hat{i} + 20\hat{j}) \text{ m/s}$$

Superposition Method

In rolling $v_C = v_{AC} = \omega R = 25 \text{ m/s}$. The superposition i.e. vector addition of the terms of equation $\vec{v}_A = \vec{v}_C + \vec{v}_{AC}$ are shown in the following figure. Resolving $v_{AC} = \omega R = 25 \text{ m/s}$ into its Cartesian components and adding to $\vec{v}_C$, we obtain

$$\vec{v}_A = \vec{v}_C + \vec{v}_{A/C} \rightarrow \vec{v}_A = 25\hat{i} + 15\hat{i} + 20\hat{j} = (40\hat{i} + 20\hat{j}) \text{ m/s}$$

Use of ICR

The contact point P is the ICR in rolling. The cylinder is in pure rotation about the ICR at the instant under consideration, so from the relative motion equation, we have

$$\vec{v}_A = \vec{\omega} \times \vec{r}_{AP} \rightarrow \vec{v}_A = \vec{\omega} \times (\vec{PC} + \vec{CA}) = (-5\hat{k}) \times \{(5\hat{j}) + (-4\hat{i} + 3\hat{j})\}$$

$$\vec{v}_A = (40\hat{i} + 20\hat{j}) \text{ m/s}$$

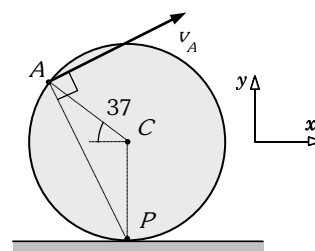
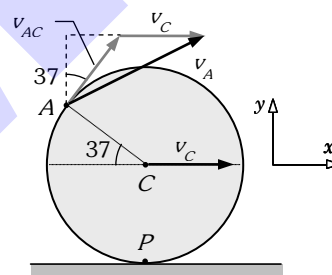
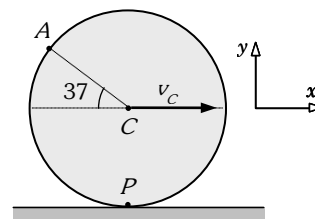
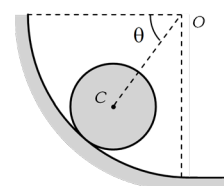


Illustration 39. A disc of radius r is rolling down a circular track of radius R .

There is no slipping between the disc and the track. When line OC is at angle θ down the horizontal, center of the disc has velocity v_C . Assume center O of the track as origin of reference frame, find angular velocity of translation motion of the center of the disc and angular velocity of rotation motion.



Solution. The angular velocity of translation motion of the center of the disc equals to the rate of change is θ . Let us denote it by ω_o .

$$\omega_o = \frac{d\theta}{dt}$$

The center of the disc moves on circular path of radius $R - r$.
Relation between velocity v_C of the center of the disc, radius R of the circular track and radius of the disc is

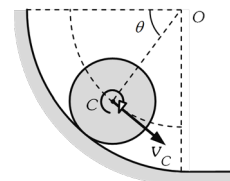
$$v_C = \omega_o(R - r)$$

Therefore, angular velocity of translation motion of center of the disc is $\omega_o = \frac{v_C}{R - r}$

Since the disc is rolling without slipping on the circular track, its angular velocity of rotation ω is given by the following equation.

$$v_C = \omega r$$

Therefore, angular velocity of rotation of the disc is $\omega = \frac{v_C}{r}$



Acceleration relations by Analytical Method

The point of contact P does not slide on the surface, therefore component of its acceleration parallel to the surface must be zero. However, it has an acceleration component towards the center. The center always moves parallel to the horizontal surface and does not change direction of its velocity; therefore, its acceleration can only be parallel to the surface.

Relation between acceleration of acceleration vector of the center C and point of contact P .

$$\vec{a}_C = \vec{a}_P + \vec{\alpha} \times \vec{PC} - \omega^2 \vec{PC} \rightarrow a_C \hat{i} = a_{Py} \hat{j} + (-\alpha \hat{k}) \times R \hat{j} - \omega^2 R \hat{j} = a_{Py} \hat{j} + \alpha R \hat{i} - \omega^2 R \hat{j}$$

Equating coefficients of x and y -components on both the sides of the above equation, we have

$$\vec{a}_C = \alpha R \hat{i}; \quad \vec{a}_P = \omega^2 R \hat{j}$$

Condition of pure rolling on fixed surface is $a_{cm} = \pm \alpha R$

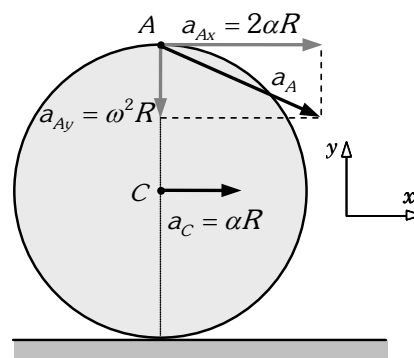
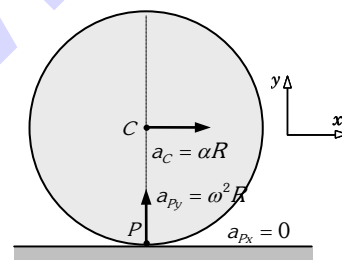
In the given figure, acceleration vectors the point of contact; center and the top point are shown. Now we will see how these accelerations can be calculated by using relative motion equation.

Once velocity of the center is obtained, we can use relative motion between A and C as well. Now we calculate acceleration of the top point A .

$$\vec{a}_A = \vec{a}_C + \vec{\alpha} \times \vec{CA} - \omega^2 \vec{CA} \rightarrow \vec{a}_A = \alpha R \hat{i} + (-\alpha \hat{k}) \times R \hat{j} - \omega^2 R \hat{j}$$

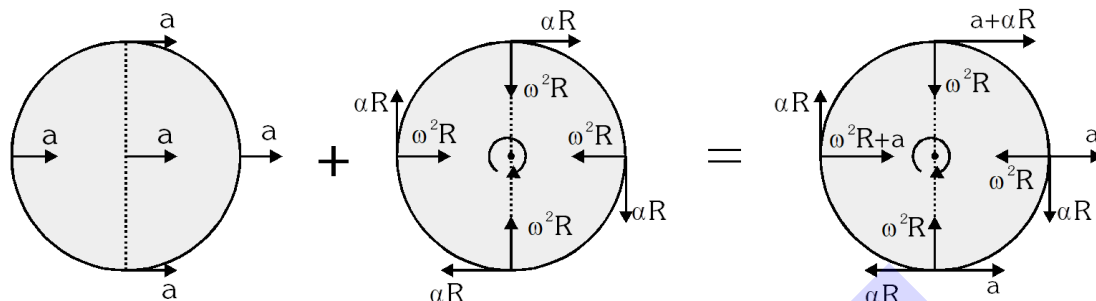
$$\vec{a}_A = 2\alpha R \hat{i} - \omega^2 R \hat{j}$$

Acceleration vector of point A and its components are shown in the given figure.



Acceleration relations by Superposition Method

Now we see how acceleration relations are expressed for a rolling wheel by assuming its rolling as superposition of its translation with the acceleration of center and simultaneous rotation about the centre.



Acceleration relations by use of ICR

Acceleration relations can also be obtained by assuming the body in pure rotation about the ICR. Here we will use relative motion equation. Always keep in mind that the acceleration of the ICR is not zero, it has value $\omega^2 R$ and points towards the center of the body. Now we will see how acceleration of the center C , the top point A and an arbitrarily chosen point B can be calculated by assuming the body in state of pure rotation about the ICR.

Acceleration of center C $\vec{a}_C = \vec{a}_P + \vec{\alpha} \times \vec{PC} - \omega^2 \vec{PC} = \alpha R \hat{i}$

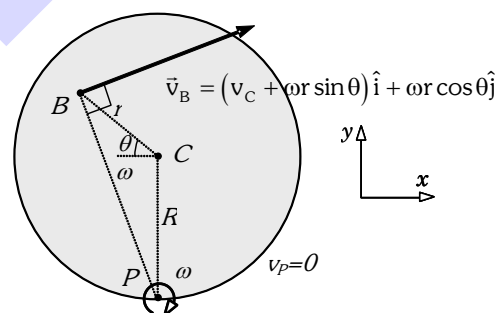
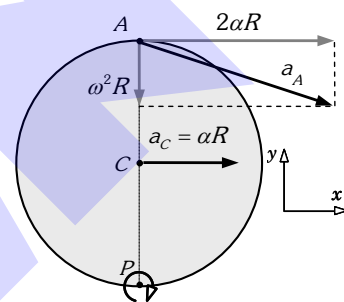
$$\vec{a}_C = \alpha R \hat{i}$$

Acceleration of the top point A $\vec{a}_A = \vec{a}_P + \vec{\alpha} \times \vec{PA} - \omega^2 \vec{PA} \hat{j}$

$$\vec{a}_A = 2\alpha R \hat{i} - \omega^2 R \hat{j}$$

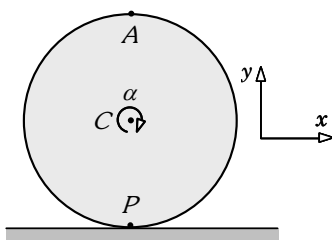
Acceleration of the point B $\vec{a}_B = \vec{a}_P + \vec{\alpha} \times \vec{PB} - \omega^2 \vec{PB}$

$$\vec{a}_B = \{\alpha(R + r \sin \theta) + \omega^2\} \hat{i} + \{\alpha r \cos \theta - \omega^2 r \sin \theta\} \hat{j}$$



Illustrations

Illustration 40. A body of round section of radius 10 cm starts rolling on a horizontal stationary surface with uniform angular acceleration 2 rad/s^2 .



(a) Find initial acceleration of the center C and top point A .

(b) Find expression for acceleration of the top point A as function of time.

Solution.

Initially when the body starts, it has no angular velocity; therefore, $\vec{a}_A = \vec{a}_B + \vec{\alpha} \times \vec{BA}$

The angular acceleration vector is $\vec{\alpha} = -2\hat{k} \text{ rad/s}^2$.

(a) Acceleration of the center C is obtained by using condition for rolling without slipping.

$$\vec{a}_C = \vec{\alpha} \times \vec{PC} \rightarrow \vec{a}_C = -2\hat{k} \times 10\hat{j} = 20\hat{i} \text{ cm/s}^2$$

Acceleration of the point A can be obtained either by analytical method, superposition method or by use of ICR. These methods for calculation of acceleration of the top point are already described; therefore, we use the result.

$$\vec{a}_A = 2\alpha R\hat{i} \rightarrow \vec{a}_A = 40\hat{i} \text{ cm/s}^2$$

(b) Initially at the instant $t = 0$, when the body starts, its angular velocity is zero. At latter time it acquires angular velocity $\vec{\omega}$, therefore acceleration of any point on the body, other than its center, has an additional component of acceleration.

Angular velocity acquired by the body at time t is

$$\vec{\omega} = \vec{\omega}_0 + \vec{\alpha}t \rightarrow \text{Substituting } \omega_0 = 0, \text{ we have } \vec{\omega} = -2t\hat{k}$$

Analytical Method

Using the relative motion equation for the pair of points C and A , we have

$$\vec{a}_A = \vec{a}_C + \vec{\alpha} \times \vec{CA} - \omega^2 \vec{CA} \rightarrow \vec{a}_A = \alpha R\hat{i} + (-\alpha\hat{k}) \times R\hat{j} - \omega^2 R\hat{j} = 2\alpha R\hat{i} - \omega^2 R\hat{j}$$

Substituting the known values

$$\vec{\alpha} = -2\hat{k} \text{ rad/s}^2, \vec{\omega} = -2t\hat{k} \text{ rad/s and } R = 10 \text{ cm}$$

$$\text{we have } \vec{a}_A = 40\hat{i} - 40t^2\hat{j} \text{ cm/s}$$

Superposition Method

We superimpose translation motion of the center and rotation motion about the center. In fact it is vector addition of terms of above equation used in analytical method.

From the above figure, we have

$$\vec{a}_A = (a_C + \alpha R)\hat{i} - \omega^2 R\hat{j}$$

Substituting known values $\vec{\alpha} = -2\hat{k} \text{ rad/s}^2, \vec{\omega} = -2t\hat{k} \text{ rad/s and } R = 10 \text{ cm},$

$$\text{we have } \vec{a}_A = 40\hat{i} - 40t^2\hat{j} \text{ cm/s}^2$$

Use of ICR

The point of contact P is the ICR, because the body is rolling without slipping. We use relative motion equation for pairs of points P and A .

$$\vec{a}_A = \vec{a}_P + \vec{\alpha} \times \vec{PA} - \omega^2 \vec{PA} \rightarrow \vec{a}_A = \omega^2 R\hat{j} + 2\alpha R\hat{i} - 2\omega^2 R\hat{j} = 2\alpha R\hat{i} - \omega^2 R\hat{j}$$

Substituting known values

$$\alpha = 2 \text{ rad/s}^2, \omega = 4t \text{ rad/s and } R = 10 \text{ cm},$$

$$\text{we have } \vec{a}_A = 40\hat{i} - 40t^2\hat{j} \text{ cm/s}^2$$

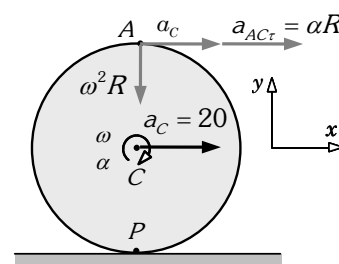
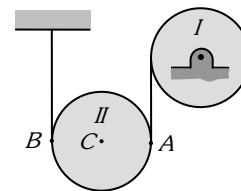


Illustration 41. Two identical discs, each of radius r , are connected by a cord as shown in the figure. The disc I rotates with constant angular acceleration α in anticlockwise direction. Find acceleration of the center of disc II and its angular acceleration.



Solution.

As the disc I rotates, the thread unwraps it. Acceleration of a point on the portion of the thread between the two discs equals to tangential acceleration of any point on the periphery of the disc I . The extreme left point A of the disc II must also descend with the same acceleration.

Downward acceleration a_B of point B = Tangential acceleration of a point on the periphery of disc I .

$$a_B = \alpha_1 r \rightarrow a_B = \alpha r$$

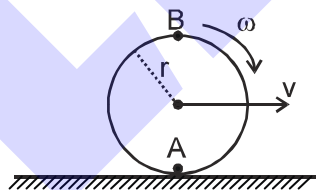
The point B on the thread is at rest relative to the ground; therefore, it can be assumed that the second disc is in a motion similar to rolling without slipping on a vertical surface.

Now applying conditions of rolling without slipping we have

$$\text{Acceleration of the center of the disc } II \quad a_C = \frac{1}{2} a_B = \frac{1}{2} \alpha r$$

$$\text{Angular acceleration of the disc } II \quad \alpha_{II} = \frac{a_C}{r} = \frac{1}{2} \alpha$$

Illustration 42. A wheel of radius r rolls (rolling without slipping) on a level road as shown in figure. Find out velocity of point A and B



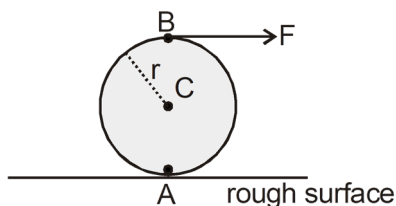
Solution

Contact surface is in rest for pure rolling velocity of point A is zero.

$$\text{so } v = \omega r$$

$$\text{velocity of point } B = v + \omega r = 2v$$

Illustration 43. A force F acts tangentially at the highest point of a sphere of mass m kept on a rough horizontal plane. If the sphere rolls without slipping, find the acceleration of the centre (C).



Solution

The situation is shown in figure. As the force F rotates the sphere, the point of contact has a tendency to slip towards left so that the static friction on the sphere will act towards right. Let r be the radius of the sphere and a be the linear acceleration of the centre of the sphere. The angular acceleration about the centre of the sphere is $\alpha = a/r$, as there is no slipping.

For the linear motion of the centre,

$$F + f = ma \quad \dots(i)$$

and for the rotational motion about the centre,

$$Fr - fr = I\alpha = \left(\frac{2}{5}mr^2\right)\left(\frac{a}{r}\right) \quad \text{or,} \quad F - f = \frac{2}{5}ma, \quad \dots(iii)$$

$$\text{From (i) and (ii), } 2F = \frac{7}{5}ma \quad \text{or} \quad a = \frac{10F}{7m}.$$

Illustration 44. A circular rigid body of mass m , radius R and radius of gyration (k) rolls without slipping on an inclined plane of a inclination θ . Find the linear acceleration of the rigid body and force of friction on it. What must be the minimum value of coefficient of friction so that rigid body may roll without sliding?

Solution

If a is the acceleration of the centre of mass of the rigid body and f the force of friction between sphere and the plane, the equation of translatory and rotatory motion of the rigid body will be

$$mg \sin \theta - f = ma \quad (\text{Translatory motion})$$

$$fR = I\alpha \quad (\text{Rotatory motion})$$

$$f = \frac{I\alpha}{R}$$

$$I = mk^2, \text{ due to pure rolling } a = \alpha R$$

$$mg \sin \theta - \frac{I\alpha}{R} = m\alpha R$$

$$mg \sin \theta = m\alpha R + \frac{I\alpha}{R}$$

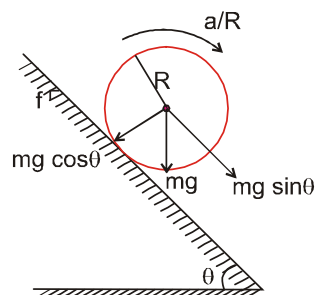
$$mg \sin \theta = m\alpha R + \frac{mk^2\alpha}{R}$$

$$mg \sin \theta = m\alpha + \frac{mk^2\alpha}{R}$$

$$mg \sin \theta = a \left[\frac{R^2 + k^2}{R^2} \right]$$

$$a = \frac{g \sin \theta}{\left[\frac{R^2 + k^2}{R^2} \right]}$$

$$f = \frac{a}{\left(1 + \frac{k^2}{R^2} \right)}$$



$$f = \frac{mk^2 a}{R^2} \Rightarrow \frac{mg k^2 \sin \theta}{R^2 + k^2}$$

$$f \leq \mu N$$

$$\frac{mk^2}{R^2} a \leq \mu \leq mg \cos \theta$$

$$R^2 \frac{k^2}{R^2} \times \frac{g \sin \theta}{(k^2 + R^2)} \leq \mu g \cos \theta$$

$$\mu \geq \left[\frac{\tan \theta}{1 + \frac{R^2}{k^2}} \right]$$

$$\mu_{\min.} = \left[\frac{\tan \theta}{1 + \frac{R^2}{k^2}} \right]$$

From above Illustration if rigid bodies are solid cylinder, hollow cylinder, solid sphere and hollow sphere.

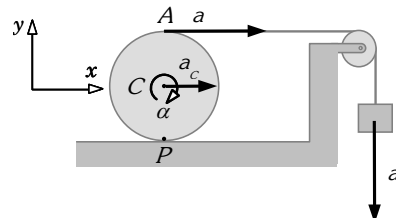
Increasing order of acceleration : $a_{\text{solid sphere}} > a_{\text{solid cylinder}} > a_{\text{hollow sphere}} > a_{\text{hollow cylinder}}$

Illustration 45. A block of mass m is attached at one end of a thin light cord, which passes over an ideal pulley. At the other end, it is wrapped around a cylinder of mass M , which can roll without slipping over a horizontal plane.

- (a) What is the acceleration of the block?
 (b) What is the friction force on the cylinder?

Solution.

The problem requires solution of the force and the torque equations consistent with the condition of rolling, so it is not necessary to decide the direction of friction. To start with let the static friction f_s acts in the forward direction.



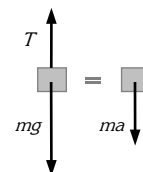
- (a) Let the block descend with acceleration a . Since the cord is inextensible the top point A of the cylinder also moves with the same acceleration. Applying relative motion equation with the condition required for rolling that the particle of the cylinder at the point of contact has no acceleration parallel to the horizontal plane.

$$\vec{a}_A = \vec{\alpha} \times \vec{r}_{A/P} \rightarrow a_A \hat{i} = (-\alpha \hat{k}) \times (2r \hat{j}) \Rightarrow \alpha = \frac{a}{2r} \quad (1)$$

From eq. (1) and relative motion equation for P and the center C , we have

$$\vec{a}_C = \vec{\alpha} \times \vec{r}_{C/P} \rightarrow a_C \hat{i} = (-\alpha \hat{k}) \times (r \hat{j}) \Rightarrow a_C = \frac{1}{2} a \quad (2)$$

The block is in translation motion under the action of its weight mg and tension T in the string. These forces and the effective force ma are shown in the adjacent figure.



Applying Newton's second law for translation of mass center, we have

$$\sum \vec{F}_i = M\vec{a}_C \rightarrow mg - T = ma \quad (3)$$

The cylinder is in rolling under the action of its weight Mg , normal reaction N from the ground; tension T in the cord and force of static friction f_s . These forces, the effective force Ma_C and the effective torque $I_C\alpha$ are shown in the adjacent figure.

Applying Newton's second law for translation of mass center, we have

$$\sum F_x = Ma_{Cx} \rightarrow T + f_s = Ma_C$$

Substituting a_C from eq. (2), we have

$$T + f_s = \frac{1}{2}Ma \quad (4)$$

Applying torque equation for centroidal rotation, we have

$$\sum \vec{\tau}_C = I_C\vec{\alpha} \rightarrow Tr - f_sr = I_C\alpha$$

Substituting $\frac{1}{2}mr^2$ for I_C , and value of α from eq. (1), we have

$$T - f_s = \frac{1}{4}Ma \quad (5)$$

From eq. (3), (4) and (5), we have

$$\text{Acceleration of the block } a = \frac{8mg}{3M + 8m}$$

(b) From eq. (4), (5) and above value of acceleration a , we have

$$\text{Force of static friction } f_s = \frac{Mmg}{3M + 8m}$$

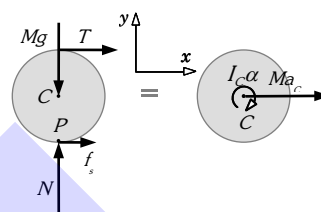
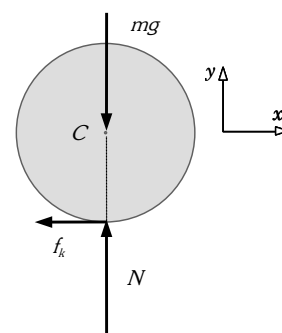
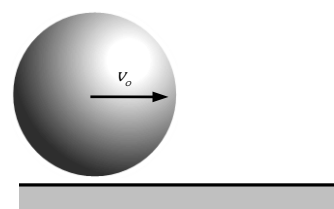


Illustration 46. A uniform sphere of mass m and radius r is projected along a rough horizontal floor with linear velocity v_0 and no angular velocity. The coefficients of kinetic and static frictions are represented by μ_s and μ_k respectively.

- How long the sphere will slide on the floor before it starts rolling.
- How far the sphere will slide on the floor before it starts rolling.
- Find the linear and angular velocities of the sphere when it starts rolling.

Solution

When the sphere touches the floor it is in translation motion. All the points including the bottom one are moving with the same velocity v_0 . Thus the bottom point which makes the contact with the floor slide on it causing kinetic friction to act in backward direction. In the adjacent figure the forces acting on the sphere are shown. Here mg represent weight, N the normal reaction from the ground and f_k .



Since the sphere has no vertical component of acceleration, by applying Newton's law we have

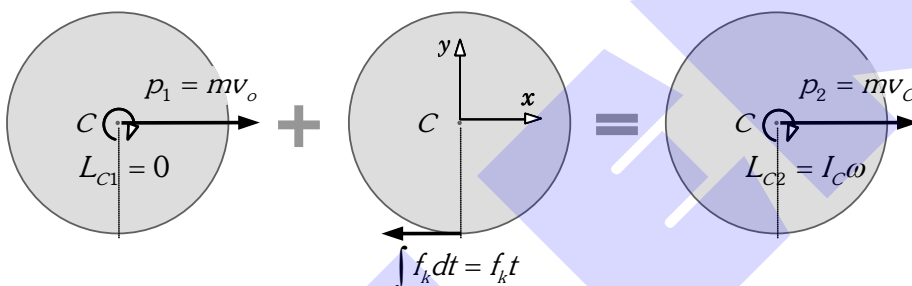
$$\sum F_y = 0 \rightarrow N = mg$$

The kinetic friction $f_k = \mu N = \mu mg$ (1)

The only force which applied torque about the centroidal axis is the kinetic friction. Angular impulse of torque of kinetic friction increases the angular velocity ω and impulse of kinetic friction decreases the mass center velocity v_C till both bear following condition required for rolling. Thereafter the sphere will continue to roll with the uniform velocity.

$$v_C = \omega r \quad (2)$$

In the adjacent figure of impulse momentum diagram the impulse of kinetic friction is shown



Translation of mass center: Applying linear impulse momentum principle in x direction, we have

$$\vec{p}_1 + \sum \vec{I}_{\text{imp}1 \rightarrow 2} = \vec{p}_2 \rightarrow p_1 - f_k t = p_2$$

Substituting p_1 , p_2 and f_k from eq. (1), we have

$$v_C = v_o - \mu g t \quad (3)$$

Centroidal rotation: Angular impulse momentum principle about the centroidal axis.

$$\vec{L}_{C1} + \sum \vec{J}_{C,1 \rightarrow 2} = \vec{L}_{C2} \rightarrow 0 + f_k r t = I_C \omega$$

Substituting $\frac{2}{5}mr^2$ for I_C and f_k from eq. (1), we have

$$\omega = \frac{5\mu g t}{2r} \quad (4)$$

(a) Substituting values of v_C and ω from eq. (3) and (4) into eq. (2), we have

$$\text{Time when rolling starts } t = \frac{2v_o}{7\mu g} \quad (5)$$

(b) Eq. (3) reveals that the mass center is in uniformly retarded motion. So its displacement in time t , when it starts rolling is given by the following equation.

$$x = \frac{1}{2}(v_o + v_C)t \rightarrow \text{Substituting values for } v_C \text{ and } t \text{ from eq. (3) and (5) respectively we have}$$

$$x = \frac{12v_o^2}{49\mu g}$$

- (c) Linear and angular velocities of the sphere when it starts rolling can be obtained by substituting t from eq. (5) into (3) and (4) respectively.

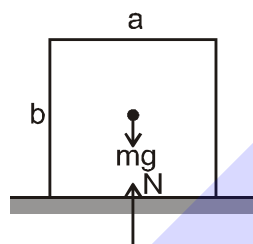
Linear velocity when rolling starts $v_C = \frac{5}{7} v_o$

Angular velocity when rolling starts $\omega = \frac{5v_o}{7r}$

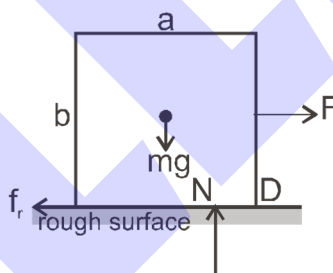
6.0 TOPPLING

In many situations an external force is applied to a body to cause it to slide along a surface. In certain cases, the body may tip over before sliding ensues. This is known as toppling.

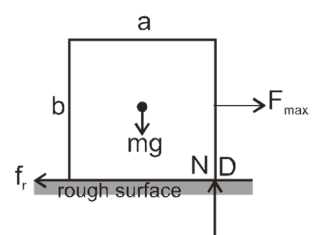
- (1) There is a no horizontal force so pressure at bottom is uniform and normal is collinear with mg .



- (2) If a force is applied at COM, pressure is not uniform Normal reaction shifts right so that torque of N can counter balance torque of friction.



- (3) If F is continuously increased N keeps shifting towards right until it reaches the right most point D . Here we have assumed that the surface is sufficiently rough so that there is no sliding as F is increase to $F_{\max}$. If force is increased any further, then torque of N can not counter balance torque of friction f_r & body will topple. The value of force now is the max value for which toppling will not occur $F_{\max}$.



$$F_{\max} = f_r$$

$$N = mg$$

torque about COM

$$f_r \cdot \frac{b}{2} = N \cdot \frac{a}{2} \Rightarrow f_r = \frac{Na}{b} = mg \frac{a}{b}, F_{\max} = mg \frac{a}{b}$$

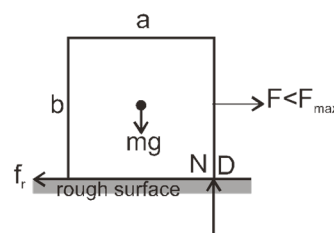
- (4) If surface is not sufficiently rough and the body slides before F is increased to $F_{\max} = mg \cdot a/b$ then body will slide before toppling. Once body starts sliding friction becomes constant and hence no toppling. This is the case if

$$F_{\max} > f_{\text{limit}} \Rightarrow mg \cdot a/b > \mu mg$$

$$\mu < a/b$$

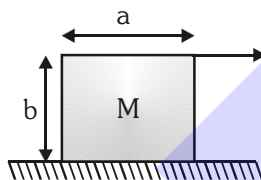
Condition for toppling when $\mu \geq a/b$ in this case body will topple if $F > mg \cdot a/b$

but if $\mu < a/b$, body will not topple any value of F applied a COM



Illustrations

Illustration 47. Find the minimum value of F to topple about an edge.



Solution

In case of toppling

Taking torque about O

$$F(b) = Mg\left(\frac{a}{2}\right) \Rightarrow F_{\min} = \frac{Mga}{2b}$$

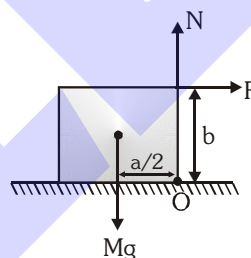


Illustration 48. A uniform cylinder of height h and radius r is placed with its circular face on a rough inclined plane and the inclination of the plane to the horizontal is gradually increased. If μ is the coefficient of friction, then under what conditions the cylinder will (a) slide before toppling (b) topple before sliding.

Solution

(a) The cylinder will slide if $mg \sin \theta > \mu mg \cos \theta \Rightarrow \tan \theta > \mu \dots (i)$

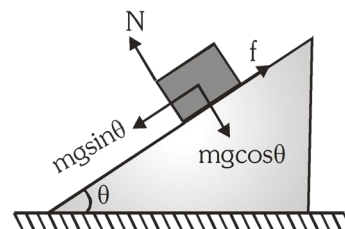
The cylinder will topple if $(mg \sin \theta) \frac{h}{2} > (mg \cos \theta) r$

$$\Rightarrow \tan \theta > \frac{2r}{h} \dots (ii)$$

Thus, the condition of sliding is $\tan \theta > \mu$ & condition of toppling is $\tan \theta > \frac{2r}{h}$.

Hence, the cylinder will slide before toppling if $\mu < \frac{2r}{h}$

(b) The cylinder will topple before sliding if $\mu > \frac{2r}{h}$



7.0 ENERGY METHODS

Newton's laws of motion tell us what is happening at an instant, while method of work and energy equips us to analyse what happens when a body moves from one place to other or a system changes its configuration. In this section, we introduce how to use methods of work and energy to analyse motion of rigid bodies.

7.1 Concept of Work in rotation motion

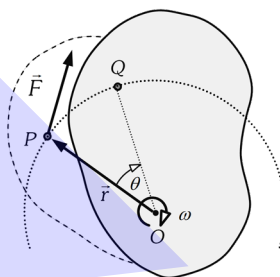
Work of a force is defined as the scalar product of the force vector and displacement vector of the point of application of the force. If during the action of a force $\vec{F}$ its point of application moves from position $\vec{r}_1$ to $\vec{r}_2$, the work $W_{1 \rightarrow 2}$ done by the force is expressed by the following equation.

$$W_{1 \rightarrow 2} = \int_{\vec{r}_1}^{\vec{r}_2} \vec{F} \cdot d\vec{r}$$

Either we can use of this idea to calculate work of a force or its modified version in terms of torque and angular displacement.

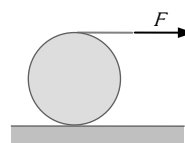
The work done by a torque during a finite rotation of the rigid body from initial value θ_i of the angle θ to final value θ_f , can be obtained by integrating both the sides of the equation given

$$W_{i \rightarrow f} = \int_{\theta_i}^{\theta_f} \vec{\tau}_o \cdot d\vec{\theta}$$

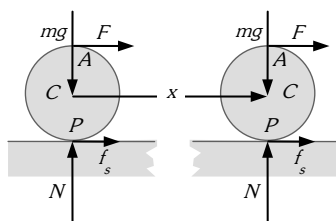


Illustrations

Illustration 49. A thin light cord is wound around a uniform cylinder placed on a rough horizontal ground. When free end of the cord is pulled by a constant force F the cylinder rolls. Denote radius of the cylinder by r and obtain expression for work done by each of the forces acting on the cylinder when center of the cylinder shifts by distance x .



Solution. Forces acting on the cylinder are its weight W , the normal reaction from the ground N , the tension T in the cord and the force of static friction f_s . The tension in the cord equals to the applied force F . These forces are shown in the adjacent figure.



In rolling point of contact P is at instantaneous rest, the center C moves with velocity $v_c = \omega r$ and the top point moves with velocity $v_A = 2v_c = 2\omega r$ both parallel to the surface on which body rolls. Since the cord is inextensible displacement of the top point equals to the displacement of the free end of the cord. These fact suggests that during displacement x of the center the free end of the cord shift through a distance $2x$.

Work done by the weight of the cylinder

$W_g = 0$ the weight is assumed to act on the center of gravity which coincides with the mass center in uniform gravitation field near the ground. The displacement x of the mass center and weight both are perpendicular to each other so the work done by gravity is zero.

Work done by the normal reaction on the cylinder

$W_N = 0$ the normal reaction acts on the particle of the body which is in contact with the ground. The particles making contact continuously change and remain at instantaneous rest during contact. Therefore normal reaction does no work.

Work done by the force of static friction.

$W_{fs} = 0$ the force of static friction f_s acts on the particle of the body which is in contact with the ground. The particles making contact continuously change and remain at instantaneous rest during contact. Therefore force of static friction f_s does no work.

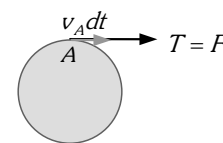
Work done by the tension in the cord

$W_T = W_F$ The particle of the wheel on which the tension in the cord acts is at the top point. Though this particle is also continuously changing but it is not in instantaneous rest and has velocity v_A . So in every infinitesimally small time interval displacement of this particle is $v_A dt = 2v_c dt$, thus work done dW_T by the tension during a time interval dt

$$dW_T = T(2v_c dt) = 2F(v_c dt)$$

When the center shifts by a distance x the work done by the tension becomes

$$W_T = W_F = 2Fx$$



7.2 Potential Energy of a rigid body

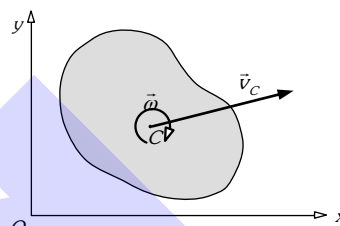
Since potential energy of a system is function of its configuration and does not depend on the manner in which the system is brought into a particular configuration, hence it does not depends on motion involved whether it is translation, rotational or their combination.

7.3 Kinetic Energy of a rigid body in rotation motion

A rigid body can be represented as a system of large number of particles, which keep their mutual distances unchanged in all circumstances. Kinetic energy of the whole body must be sum of kinetic energies of all of its particles. In this section we develop expressions for kinetic energy of a rigid body.

Kinetic Energy of a rigid body in plane motion

In the figure shown a body is in plane motion. Its mass center at an instant is moving with velocity $\vec{v}_C$ and rotating with angular velocity $\vec{\omega}$. Both these motions are shown superimposed in the given figure.



Kinetic energy too can be written as sum of kinetic energy ($\frac{1}{2}Mv_C^2$) due to translation motion of the mass center and kinetic energy ($\frac{1}{2}I_C\omega^2$) due to centroidal rotation.

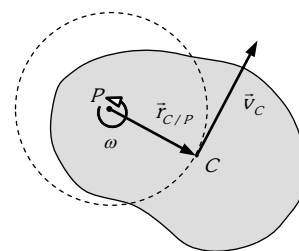
$$K = \frac{1}{2}Mv_C^2 + \frac{1}{2}I_C\omega^2$$

If location of the instantaneous axis of rotation (IAR) is known, making use of the parallel axis theorem we can write kinetic energy by the following equation also.

$$K = \frac{1}{2}I_{IAR}\omega^2$$

Kinetic Energy of a rigid body in rotation about fixed axis not passing through the mass centre

In this kind of motion the mass center is in circular motion about the axis of rotation. In the figure is shown a body rotation with angular velocity ω about a fixed axis through point P and perpendicular to plane of the paper. Mass center moves with speed $v_C = \omega r$. Kinetic energy of the body can now be expressed by the following equation.



$$K = \frac{1}{2}Mv_C^2 + \frac{1}{2}I_C\omega^2$$

Making use of the parallel axis theorem ($I_P = Mr_{P/C}^2 + I_C$) we can write kinetic energy by the following equation also.

$$K = \frac{1}{2}I_P\omega^2$$

Kinetic Energy of a rigid body in pure centroidal rotation

In pure centroidal rotation the mass center remain at rest; therefore kinetic energy due to translation of mass center vanishes.

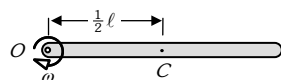
$$K = \frac{1}{2}I_C\omega^2$$

Illustrations

Illustration 50. A rod of mass m and length ℓ is pivoted to a fixed support at one of its ends O . It is rotating with constant angular velocity ω . Write expression for its kinetic energy.



Solution If the point C is the mass center of the rod, from theorem of parallel axes, the moment of inertia I_O of the rod about the fixed axis is



$$I_O = I_C + m(OC)^2 \rightarrow I_O = I_C + \frac{1}{4}m\ell^2$$

Substituting $\frac{1}{12}m\ell^2$ for I_C , we have

$$I_O = \frac{1}{3}m\ell^2$$

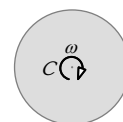
Kinetic energy of the rod equals to kinetic energy due to rotation about the fixed axis.

$$K = \frac{1}{2}I_O\omega^2 \rightarrow \text{Using above expression for } I_O, \text{ we have}$$

$$K = \frac{1}{6}m\ell^2\omega^2$$

Illustration 51. A uniform rigid body of mass m and round section of radius r is rolling on horizontal ground with angular velocity ω . Its radius of gyration about the centroidal axis is k .

- (a) Write expression of its kinetic energy.
 (b) Also express the kinetic energy as sum of kinetic energy due to translation of mass center and kinetic energy due to simultaneous centroidal rotation.



Solution (a) The point of contact with ground of a body rolling on the ground is its ICR. Let the point P is the ICR as shown in the adjacent figure. The geometrical center C of a uniform body and the mass center coincide. Therefore moment of inertia I_P of the body about the ICR can be written by using the theorem of parallel axes

$$I_P = I_C + m(PC)^2 \rightarrow I_P = I_C + mr^2$$

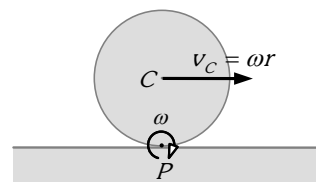
Substituting $I_C = mk^2$, we have

$$I_P = m(k^2 + r^2) \quad (1)$$

Kinetic energy of a rigid body equals to kinetic energy due to rotation about the ICR.

$$K = \frac{1}{2}I_P\omega^2 \rightarrow \text{Substituting } I_P \text{ from eq. (1), we have}$$

$$K = \frac{1}{2}m(k^2 + r^2)\omega^2$$



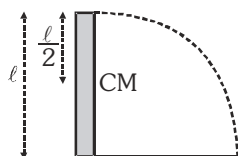
- (b) Kinetic energy of the body also equals to sum of kinetic energy due to translation of its mass center and kinetic energy due to simultaneous centroidal rotation.

$K = \frac{1}{2}mv_C^2 + \frac{1}{2}I_C\omega^2 \rightarrow$ Substituting condition for rolling $v_C = \omega r$ and $I_C = mk^2$, we have

$$K = \frac{1}{2}m(\omega r)^2 + \frac{1}{2}mk^2\omega^2 = \frac{1}{2}m(r^2 + k^2)\omega^2$$

Illustration 52. A thin meter scale is kept vertical by placing its one end on floor, keeping the end in contact stationary, it is allowed to fall. Calculate the velocity of its upper end when it hit the floor.

Solution



Loss in PE = gain in rotational

$$KE = \frac{mg\ell}{2} = \frac{1}{2}I\omega^2 = \frac{1}{2} \frac{m\ell^2}{3} \times \frac{v^2}{\ell^2} \Rightarrow v = \sqrt{3g\ell}$$

7.4 Power

Power defined as the time rate of work done, takes into account the duration in which work is done. To calculate power we make use of the following equation.

$$P = \frac{dW}{dt}$$

Instantaneous power of a force can be expressed by the following equation. Here velocity $\vec{v}$ is the velocity of the point of application of the force $\vec{F}$.

$$P = \vec{F} \cdot \vec{v}$$

7.5 Work and Energy Theorem

Work energy theorem can be applied in similar fashion as it was applied to analyze translation motion of a single body or a system of several bodies.

The work energy theorem relates kinetic energy K_1 and K_2 of a body in its initial and final position with work $W_{1 \rightarrow 2}$ done by all the external forces acting on the body to carry it from the initial position to the final position according to the following equation.

$$W_{1 \rightarrow 2} = K_2 - K_1$$

This equation is applicable in all inertial as well as noninertial frames. To write equation of work energy theorem kinetic energy must be written relative to the frame under consideration. To calculate work consider only all the physical forces in inertial frame and all the physical forces as well as pseudo force in non-inertial frame and displacement of point of applications of these forces relative to the frame under consideration.

For a system of several bodies the corresponding equation of work energy theorem can be obtained by applying the theorem for each individual body and then adding all of them. In this way we obtain an equation of the following form.

$$\Sigma W_{1 \rightarrow 2} = K_{2,s} - K_{1,s}$$

Here the term $\Sigma W_{1 \rightarrow 2}$ equals to the work of all the forces acting on various bodies irrespective of whether the force are internal or external from point of view of the system under consideration. In systems of several bodies interconnected by links of constant length e.g. inextensible cords, rods etc or body in direct contact the total work of internal forces vanishes. The work done by internal conservative forces can be accounted for by decrease in corresponding potential energies. The terms $K_{2,s}$ and $K_{1,s}$ are total kinetic energies of all the bodies in initial and final configurations of the system.

7.6 Conservation of Mechanical Energy

The work of conservative forces equals to decrease in potential energy. When a single rigid body moves or a system of rigid body changes its configuration under the action of conservative forces and nonconservative forces are either not present or if present do no work, the work energy principle can be expressed as

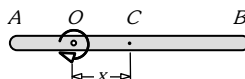
$$U_{1,s} + K_{1,s} = U_{2,s} + K_{2,s}$$

The above equation expresses the law of conservation of mechanical energy and states that if a rigid body moves or a system consisting of several rigid bodies changes its configuration under action of conservative forces the mechanical energy i.e. sum of kinetic and potential energy remain constant; provided that non-conservative, if present, do no work,

Though the work energy principle and the law of conservation of mechanical energy are equivalent, we prefer to use the former to account for non-conservative forces easily

Illustrations

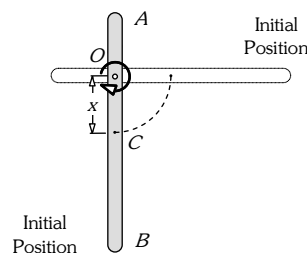
Illustration 53. A uniform rod AB of mass m and length ℓ is pivoted at a point (O) to rotate in the vertical plane. The rod is held in horizontal position and released. Find the distance x of the pivot from the mass center (C) of the rod, so that angular speed ω of the rod as it passes through the vertical position is maximum.



Solution.

The problem involves change in angular velocity with change in position, therefore demands application of principle of work and energy.

The rod when released rotates about a fixed horizontal axis passing through the point O. Its initial and final positions are shown in the adjacent figure.



Moment of inertia of the rod about the pivot O can be calculated by theorem of parallel axes.

$I_o = I_C + mx^2 \rightarrow$ Substituting $\frac{1}{12}m\ell^2$ for I_C , we have

$$I_o = \frac{1}{12}m(\ell^2 + 12x^2) \quad (1)$$

Kinetic energy in the initial position.

$$K = \frac{1}{2}I_o\omega^2 \rightarrow K_1 = 0 \quad (2)$$

Kinetic energy in the final position.

$K = \frac{1}{2}I_o\omega^2$ Substituting for I_o from eq. (1), we have

$$K_2 = \frac{1}{24}m(\ell^2 + 12x^2)\omega^2 \quad (3)$$

Only gravity does work when the rod moves from the initial to final position.

$$W = \int \vec{F} \cdot d\vec{r} \rightarrow W_{1 \rightarrow 2} = mgx \quad (4)$$

Substituting values from eq. (2), (3) and (4) in equation of work energy principle, we have

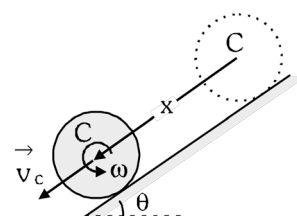
$$W_{1 \rightarrow 2} = K_2 - K_1 \rightarrow \omega = \sqrt{24gx/(\ell^2 + 12x^2)} \quad (5)$$

The above equation expresses angular velocity of the rod when it passes the vertical position. For it to be maximum

$$\frac{d\omega}{dx} = 0 \rightarrow x = \frac{\ell}{\sqrt{12}}$$

Illustration 54. A uniform rigid body of mass m and round section of radius r rolls down a slope inclined at an angle θ to the horizontal. The radius of gyration of the body about its central axis of symmetry is k .

- Derive suitable expressions for angular velocity and velocity of its mass center after it covers a distance x .
- Obtain expression for its angular acceleration and acceleration of the mass center.



Solution

The problem involves change in angular velocity with change in position, therefore demands application of principle of work and energy

The geometrical center and mass center for uniform bodies coincide; therefore center C is the mass center.

In rolling the point of contact P must always be at instantaneous rest and angular velocity ω , velocity of center C, angular acceleration α and acceleration of the center must bear the following relations.

$$v_c = \omega r \text{ and } a_c = \alpha r \quad (1)$$

(a) The rolling motion can be analyzed as superposition of translation of the mass center and simultaneous centroidal rotation.

Kinetic energy in the initial position.

$$K_1 = 0 \quad (2)$$

Kinetic energy in the final position.

$K = \frac{1}{2}mv_C^2 + \frac{1}{2}I_C\omega^2 \rightarrow$ Substituting for $I_C = mk^2$ and v_C from eq. (1), we have

$$K_2 = \frac{1}{2}m(r^2 + k^2)\omega^2 \quad (3)$$

The forces acting on the body are its weight mg , the normal reaction N from the slope and the force of static friction f_s . These forces and displacement are shown in the adjacent figure. The normal reaction and the force of static friction do no work in rolling, it is the weight, which does work.

$$W = \int \vec{F} \cdot d\vec{r} \rightarrow W_{1 \rightarrow 2} = mgx \sin \theta \quad (4)$$

Substituting values from eq. (2), (3) and (4) in equation of work energy principle, we have

$$W_{1 \rightarrow 2} = K_2 - K_1 \rightarrow \omega = \sqrt{\frac{2gx \sin \theta}{k^2 + r^2}}$$

Substituting v_C from eq. (1), we have

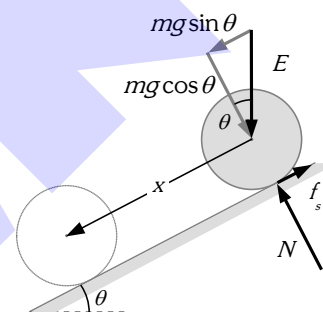
$$v_C = \sqrt{\frac{2gx \sin \theta}{(k^2/r^2) + 1}}$$

Acceleration a , velocity v and position coordinate x bear the relation $a = v(dv/dx)$. Therefore acceleration of mass center of the body.

$$a_C = v_C \frac{dv_C}{dx} \rightarrow a_C = \frac{g \sin \theta}{(k^2/r^2) + 1}$$

Substituting a_C in eq. (1), we have the angular acceleration of the body.

$$a_C = \alpha r \rightarrow \alpha = \frac{g \sin \theta}{(k^2/r) + r}$$



8.0 METHODS OF IMPULSE AND MOMENTUM

Methods of impulse and momentum describe what happens over a time interval. When motion of a body involves rotation we have to consider angular impulse as well as angular momentum. In this section we discuss concept of angular impulse, angular momentum of rigid body, angular impulse momentum principle and conservation of angular momentum.

8.1 Angular Impulse

Like impulse of a force angular impulse of a constant torque equals to product of the torque and concerned time interval and if the torque is not constant it must be integrated with time over the concerned time interval.

If torque $\vec{\tau}_o$ about an axis passing through O is constant, its angular impulse during a time interval from t_1 to t_2 denoted by $\vec{J}_{o,1 \rightarrow 2}$ is given by the following equation.

$$\vec{J}_{o,1 \rightarrow 2} = \vec{\tau}_o (t_2 - t_1)$$

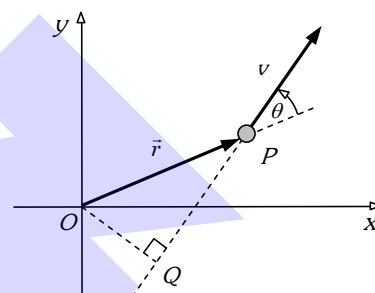
If torque $\vec{\tau}_o$ about an axis passing through O is time varying, its angular impulse during a time interval from t_1 to t_2 denoted by $\vec{J}_{o,1 \rightarrow 2}$ is given by the following equation.

$$\vec{J}_{o,1 \rightarrow 2} = \int_{t_1}^{t_2} \vec{\tau}_o dt$$

8.2 Angular momentum of a particle

Angular momentum $\vec{L}_o$ about the origin O of a particle of mass m moving with velocity $\vec{v}$ is defined as the moment of its linear momentum $\vec{p} = m\vec{v}$ about the point O.

$$\vec{L}_o = \vec{r} \times (m\vec{v})$$



8.3 Angular Momentum of a Rigid Body

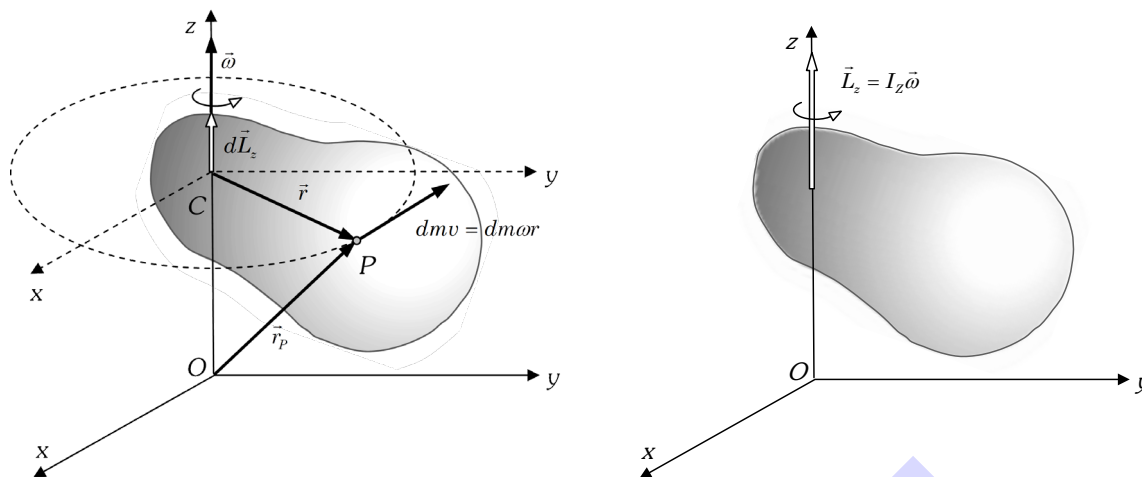
Angular momentum is quantity of rotation motion in a body. The angular momentum of a system of particles is the sum of angular momentum all the particles within the system. A rigid body is an assemblage of large number of particles maintaining their mutual distances intact under all circumstances, therefore angular momentum of a rigid body must be sum of angular momenta of all of its particles.

Angular Momentum about a point and about an axis

Angular momentum of a particle is not defined about an axis instead it is defined about a point. Therefore above idea of summing up angular momenta of all the particles about a point gives angular momentum of the rigid body about a point. But while dealing with fixed axis rotation or rotation about axis in translation we need angular momentum about an axis.

Angular momentum about an axis is calculated similar to torque about an axis. To calculate angular momentum of a particle of rigid body about an axis we take moment of momentum of the particle about the point where plane of motion of the point of application of the force intersects the axis.

In the following figure is shown angular momentum $d\vec{L}_z = \vec{r} \times (dm\vec{v}) = r^2 dm \omega$ of a particle P of a rigid body rotating about the z-axis. It is along the z-axis i.e. axis of rotation. In the next figure total angular momentum $\vec{L}_z = \int d\vec{L}_z = I_z \omega$ about the axis of rotation is shown. It is also along the axis of rotation.



Angular Momentum in general plane motion

Angular momentum of a body in plane motion can also be written similar to torque equation or kinetic energy as sum of angular momentum about the axis due to translation of mass center and angular momentum of centroidal rotation about centroidal axis parallel to the original axis.

Consider a rigid body of mass M in plane motion. At the instant shown its mass center has velocity $\vec{v}$ and it is rotating with angular velocity $\vec{\omega}$ about an axis perpendicular to the plane of the figure. Its angular momentum $\vec{L}_O$ about an axis passing through the origin and parallel to the original is expressed by the following equation.

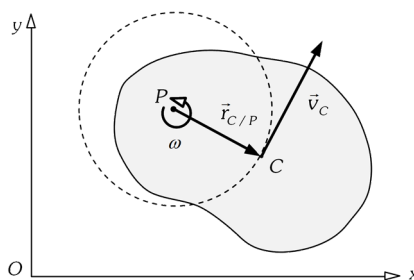
$$\vec{L}_O = \vec{r}_C \times (M\vec{v}_C) + I_C \vec{\omega}$$

The first term of the above equation represents angular momentum due to translation of the mass center and the second term represents angular momentum in centroidal rotation.

Angular momentum in rotation about fixed axis

Consider a body of mass M rotating with angular velocity ω about a fixed axis perpendicular to plane of the figure passing through point P . Making use of the parallel axis theorem $I_P = Mr_{C/P}^2 + I_C$ and equation $\vec{v}_C = \vec{\omega} \times \vec{r}_{C/P}$ we can express the angular momentum $\vec{L}_P$ of the body about the fixed rotational axis.

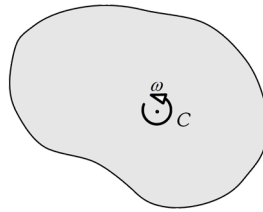
$$\vec{L}_P = I_P \vec{\omega}$$



The above equation reveals that the angular momentum of a rigid body in plane motion can also be expressed in a single term due to rotation about the instantaneous axis of rotation.

Angular momentum in pure centroidal rotation

In pure centroidal rotation, mass center remains at rest, therefore angular momentum due to translation of the mass center vanishes.



$$\vec{L}_C = I_C \vec{\omega}$$

Rotational Equivalent of the Newton's Laws of Motion

Differentiating terms on both the sides of equation $\vec{L}_O = \vec{r}_C \times (M\vec{v}_C) + I_C \vec{\omega}$ with respect to time, and making substitution of $\vec{v}_C = d\vec{r}_C/dt$, $\vec{a}_C = d\vec{v}_C/dt$ and $\vec{\alpha} = d\vec{\omega}/dt$ we have

$$\frac{d\vec{L}_O}{dt} = \vec{v}_C \times (M\vec{v}_C) + \vec{r}_C \times M\vec{a}_C + I_C \vec{\alpha}$$

The first term on the right hand side vanishes, so we can write

$$\frac{d\vec{L}_O}{dt} = \vec{r}_C \times M\vec{a}_C + I_C \vec{\alpha}$$

Now comparing the above equation with torque equation $\Sigma \vec{\tau}_O = \vec{r}_C \times M\vec{a}_C + I_C \vec{\alpha}$, we have

$$\Sigma \vec{\tau}_O = \frac{d\vec{L}_O}{dt}$$

The above equation though developed for plane motion only yet is valid for rotation about an axis in rotation also. It states that the net torque about the origin of an inertial frame equals to the time rate of change in angular momentum about the origin and can be treated as a parallel to Newton's second law which states that net external force on a body equals to time rate of change in its linear momentum.

Angular Impulse Momentum Principle

Rearranging the terms and integrating both the sides obtained from previous equation, we can write

$$\Sigma \int_{t_1}^{t_2} \vec{\tau}_O dt = \vec{L}_{O2} - \vec{L}_{O1}$$

The left hand side of the above equation is the angular impulse of torque of all the external forces in the time interval in the time interval t_1 to t_2 .

$$\Sigma \vec{J}_{O,1 \rightarrow 2} = \vec{L}_{O2} - \vec{L}_{O1}$$

The idea expressed by the above equation is known as angular impulse momentum principle and states that increment in the angular momentum of a body about a point in a time interval equals to the net angular impulse of all the external forces acting on it during the concerned time interval.

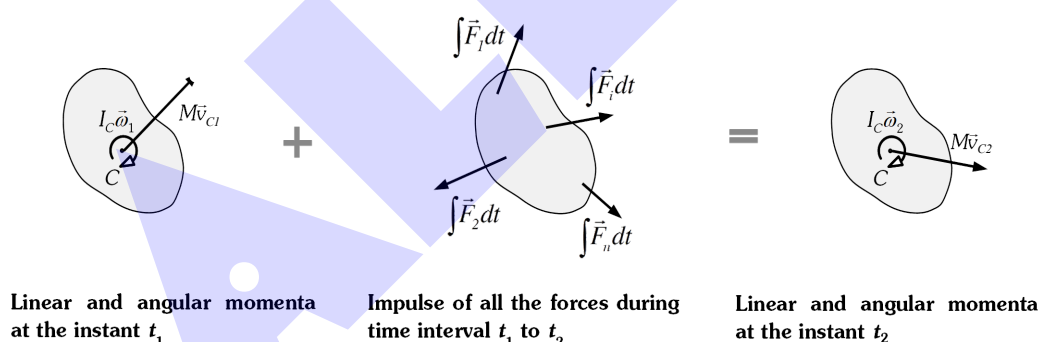
For the ease of application the above equation is rearranged as

$$\vec{L}_{o1} + \sum \vec{J}_{o,1 \rightarrow 2} = \vec{L}_{o2}$$

Like linear impulse momentum principle, the angular impulse momentum principle provides us solution of problems concerned with change in angular velocity in a time interval or change in angular velocity during very short interval interactions.

Method of Impulse Momentum Principle for Plane motion of a Rigid Body

Linear momentum and angular momentum serve as measures of amount of translation and rotation motion respectively. The external forces acting on a rigid body can change its state of translation as well as rotation motion which is reflected by change in linear as well as angular momentum according to the principles of linear impulse and momentum and angular impulse and momentum.



In the above figure is shown strategy to apply method of impulse and momentum. Consider a rigid body of mass M in plane motion. Its moment of inertia about the centroidal axis perpendicular to plane of motion is I_C . Let $\vec{v}_{C1}$ and $\vec{\omega}_1$ represent velocity of its mass center and its angular velocity at the beginning of a time interval t_1 to t_2 . Under the action of several forces $\vec{F}_1, \vec{F}_2 \dots \vec{F}_1 \dots \vec{F}_n$ during the time interval its mass center velocity and angular velocity become $\vec{v}_{C2}$ and $\vec{\omega}_2$ respectively. The adjacent figure shows strategy representing how to write equations for linear and angular impulse momentum principles.

While applying the principle it becomes simpler to consider translation of the mass center and centroidal rotation separately. Thus in an alternative way we apply linear impulse momentum principle for translation of the mass center and angular impulse momentum principle for centroidal rotation.

Translation of mass center: Linear impulse momentum principle.

$$\vec{p}_1 + \sum \vec{I}_{mp1 \rightarrow 2} = \vec{p}_2$$

Here $\vec{p}_1 = M\vec{v}_{C1}$ and $\vec{p}_2 = M\vec{v}_{C2}$ represent linear momentums at the beginning and end of the time interval and $\sum \vec{I}_{mp1 \rightarrow 2}$ stands for impulse of all the external forces during the time interval.

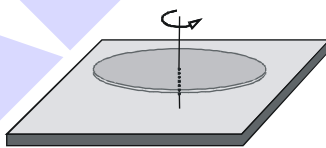
Centroidal rotation: Angular impulse momentum principle.

$$\vec{L}_{C1} + \sum \vec{J}_{C,1 \rightarrow 2} = \vec{L}_{C2}$$

Here $\vec{L}_{C1} = I_C \vec{\omega}_1$ and $\vec{L}_{C2} = I_C \vec{\omega}_2$ represent angular momentums about the centroidal axis at the beginning and end of the time interval and $\sum \vec{J}_{C,1 \rightarrow 2}$ stands for angular impulse of all the external forces about the centroidal axis during the time interval.

Illustrations

Illustration 55. A uniform disc of mass M and radius R rotating with angular velocity ω_0 about a vertical axis passing through its center and perpendicular to its plane is placed gently on a rough horizontal ground, where coefficient of friction is μ . How long it will take to stop.



Solution The torque of friction forces is

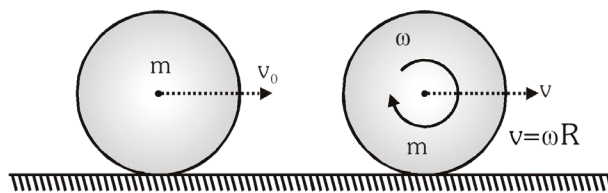
$$\tau_C = \frac{2}{3} \mu MgR \quad (1)$$

The angular impulse of the torque of friction is responsible to stop the disc. Applying angular impulse momentum principle, we have

$$\vec{L}_{C1} + \sum \vec{J}_{C,1 \rightarrow 2} = \vec{L}_{C2} \rightarrow I_C \omega_0 - \tau_C t = 0$$

Substituting $I_C = \frac{1}{2}MR^2$ and τ_C from eq. (1), we have $t = \frac{3R\omega_0}{4\mu g}$

Illustration 56. A body of radius R and mass m is placed on horizontal rough surface with linear velocity v_0 , after some time it comes in the condition of pure rolling then determine:



- Time t at which body starts pure rolling.
- Linear velocity of body at time t .
- Work done by frictional force in this time t .

Solution

For translatory motion $v = u + at$

Initial velocity $u = v_0$

Let after time t pure rolling starts and at this time t final velocity $= v$ and acceleration $= a$

From FBD :

Normal Reaction $N = mg$

Friction force $f = \mu N = \mu mg \Rightarrow ma = \mu mg$ [$\because f = ma$]

Retardation $a = \mu g$

$$v = v_0 - at \text{ (-ve sign for retardation)}$$

$$v = v_0 - \mu gt \quad \dots(i)$$

For rotatory motion $\omega = \omega_0 + \alpha t$ (Initial angular velocity $\omega_0 = 0$)

$$\Rightarrow \omega = \alpha t \quad \dots(ii)$$

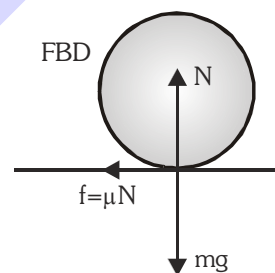
$$\because \tau = I\alpha \Rightarrow \alpha = \frac{\tau}{I} = \frac{fR}{mK^2} = \frac{\mu mgR}{mK^2} = \alpha = \frac{\mu gR}{K^2} \quad \dots(iii)$$

$$\text{From eq}^n. (ii) \text{ and eq}^n. (iii) \quad \omega = \frac{\mu gR}{K^2} t \quad \dots(iv)$$

$$\because \text{For pure rolling } v = \omega R \Rightarrow \omega = \frac{v}{R} \quad \dots(v)$$

$$\text{From eq. (iv) and (v)} \quad \frac{v}{R} = \frac{\mu gR}{K^2} t \text{ or } v = \frac{\mu gR^2 t}{K^2} \quad \dots(vi)$$

$$\text{substitute } v \text{ from eq}^n. (vi) \text{ into eq}^n. (i) \quad \frac{\mu gR^2 t}{K^2} = v_0 - \mu gt \Rightarrow t = \frac{v_0}{\mu g \left[1 + \frac{R^2}{K^2} \right]}$$



Putting the value of t in equation (i) $v = v_0 - \mu g \frac{v_0}{\mu g \left[1 + \frac{R^2}{K^2} \right]} = v_0 - \frac{v_0}{1 + \frac{R^2}{K^2}} = \frac{v_0}{1 + \frac{K^2}{R^2}}$

Work done in sliding by frictional force = Initial kinetic energy – Final kinetic energy

Work done by friction $W_f = \frac{1}{2} M v_0^2 - \frac{1}{2} M v^2 \left(1 + \frac{K^2}{R^2} \right) = \frac{1}{2} M v_0^2 - \frac{1}{2} \frac{M v_0^2}{\left(1 + \frac{K^2}{R^2} \right)} = \frac{M v_0^2}{2 \left(1 + \frac{R^2}{K^2} \right)}$

8.4 Conservation of Angular Momentum

If angular impulse of all the external forces about an axis in time interval vanishes, the angular momentum of the system about the same axis in that time interval remain unchanged.

If $\sum \int_{t_1}^{t_2} \vec{\tau}_o dt = 0$, we have $\vec{L}_{o1} = \vec{L}_{o2}$

The condition of zero net angular impulse required for conservation of angular momentum can be fulfilled in the following cases.

- If no external force acts, the angular impulse about all axes will be zero and hence angular momentum remains conserved about all axes.
- If net torque of all the external forces or torques of each individual force about an axis vanishes the angular momentum about that axes will be conserved.
- If all the external forces are finite in magnitude and the concerned time interval is infinitely small, the angular momentum remain conserved.
- If a system of rigid bodies changes its moment of inertia by changing its configuration due to internal forces only its angular momentum about any axes remains conserved. If we denote the moment of inertias in two configurations by I_1 and I_2 and angular velocities by ω_1 and ω_2 , we can write

$$I_1 \vec{\omega}_1 = I_2 \vec{\omega}_2$$

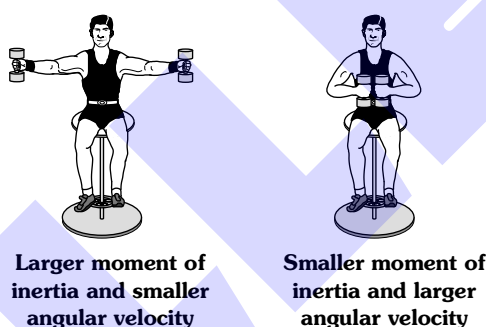
The principle of conservation of angular momentum governs a wide range of physical processes from subatomic to celestial world. The following Illustration s explicate some of these applications.

Spinning Ice Skater

A spinning ice skater and ballet dancers can control her moment of inertia by spreading or bringing closer her hands and make use of conservation of angular momentum to perform their spins. In doing so no external forces is needed and if we ignore effects of friction from the ground and the air, the angular momentum can be assumed conserved. When she spreads her hand or leg away, her moment of inertia decreases therefore her angular velocity decreases and when she brings her hands or leg closer her moment of inertia increases therefore her angular velocity increases.

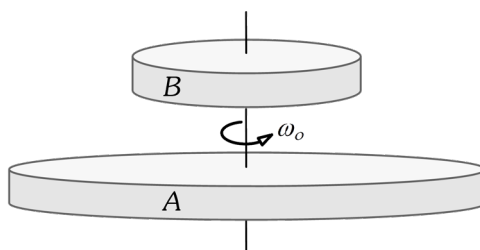
Student on rotating turntable

The student, the turntable and dumbbells make an isolated system on which no external torque acts, if we ignore friction in the bearing of the turntable and air friction. Initially the student has his arm stretched on rotating turntable. When he pulls dumbbells close to his body, angular velocity increases due to conservation of angular momentum.



Illustrations

Illustration 57. Consider the disc A of moment of inertia I_1 rotating freely in horizontal plane about its axis of symmetry with angular velocity ω_0 . Another disc B of moment of inertia I_2 held at rest above the disc A. The axis of symmetry of the disc B coincides with that of the disc A as shown in the figure. The disc B is released to land on the disc A. When sliding stops, what will be the angular velocity of both the discs?



Solution

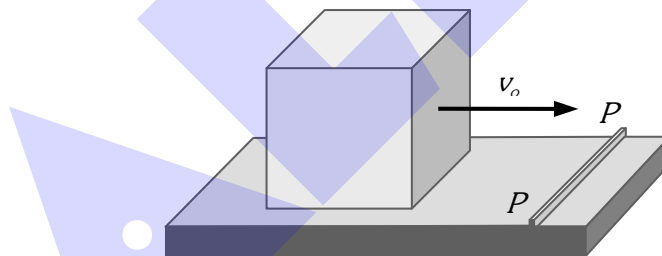
Both the discs are symmetric about the axis of rotation therefore does not require any external torque to keep the axis stationary. When the disc B lands on A slipping starts. The force of friction provides an internal torque to system of both the disc. It slows down rotation rate of A and increases that of B till both acquire same angular velocity ω .

Since there is no external torques on the system of both the discs about the axis of rotation, the total angular momentum of the system remains conserved. The total angular momentum of the system is the sum of angular momentum of both discs. Denoting the angular momentum of the disc A before B lands on it and long after slipping between them stops by symbols $\vec{L}_{A1}$, $\vec{L}_{B1}$, $\vec{L}_{A2}$ and $\vec{L}_{B2}$ respectively, we can express conservation of angular momentum by the following equation.

$$\vec{L}_{A1} + \vec{L}_{B1} = \vec{L}_{A2} + \vec{L}_{B2} \rightarrow I_1\omega_o + 0 = I_1\omega + I_2\omega \Rightarrow \omega = \frac{I_1\omega_o}{I_1 + I_2}$$

Illustration 58. A cube of mass m and edge length ℓ can slide freely on a smooth horizontal floor.

Moving on the floor with velocity v_o , it strikes a long obstruction PP of small height. The obstruction is parallel to the leading bottom edge of the cube. The leading bottom edge gets pivoted with the obstruction and the cube starts rotating. Determine angular velocity of the cube immediately after the impact.

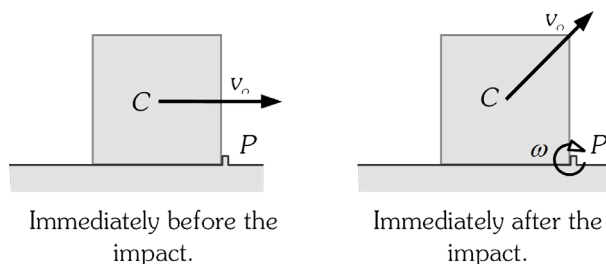


Solution

Before the impact, there is no external force in the horizontal direction and the cube slides with uniform velocity v_o , and during the impact reaction forces of the obstruction stops its leading bottom edge and cause it to rotate about its leading bottom.

During the impact external forces acting on the cube are its weight, the normal reaction from the ground and reaction from the obstruction. The weight and the normal reaction from the ground both are finite in magnitude and the impact ends in infinitesimally small time interval so their impulses and angular impulses about any axes are negligible. It is the reaction from the obstruction which has finite impulse during the impact.

Its horizontal component changes the momentum of the cube during the impact, but its angular impulse about the obstruction is zero, therefore the angular momentum of the cube about an axis coincident with the leading bottom edge remain conserved.



Let the velocity of the mass center and angular velocity of the cube immediately after the impact are v_{C_o} and ω_o . These velocities are shown in the adjacent figure.

We denote the angular momentum of the cube about axis coincident with the obstruction edge before and after the impact by L_{P1} and L_{P2} .

Applying principle of conservation of angular momentum about an axis coincident with the obstruction, we have

$$\vec{L}_{P1} = \vec{L}_{P2} \rightarrow \vec{r}_{C/P} \times m\vec{v}_o = I_P \vec{\omega}$$

Using theorem of parallel axes for moment of inertia I_P about the leading bottom edge, we get. $I_P = I_C + mr_{C/P}^2 = \frac{2}{3}m\ell^2$ Substituting this in the above equation, we have

$$\text{Angular velocity immediately after the impact } \omega_o = \frac{3v_o}{4\ell}$$

Angular momentum of a body in combined translational and rotational motion

Suppose a body is rotating about an axis passing through its centre of mass with an angular velocity ω_{cm} and moving translationally with a linear velocity v . Then, the angular momentum of the body about a point P outside the body in the lab frame is given by, $\vec{L}_P = \vec{L}_{cm} + \vec{r} \times \vec{p}_{cm}$ where $\vec{r}$ is the position vector of the centre of mass with respect to point P. Hence, $\vec{L}_P = I_{cm} \vec{\omega} + \vec{r} \times m\vec{v}_{cm}$

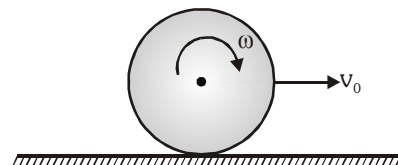
Illustrations

Illustration 59. A solid sphere rolls without slipping on a rough surface and the centre of mass has constant speed v_o . If mass of the sphere is m and its radius is R , then find the angular momentum of the sphere about the point of contact.

Solution $\therefore \vec{L}_P = \vec{L}_{cm} + \vec{r} \times \vec{p}_{cm} = I_{cm} \vec{\omega} + \vec{R} \times m\vec{v}_{cm}$

Since sphere is in pure rolling motion hence

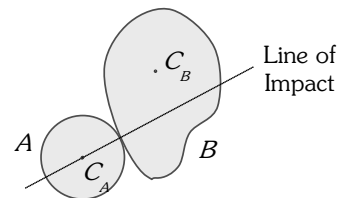
$$\omega = v_o R \Rightarrow \vec{L}_P = \left(\frac{2}{5} MR^2 \frac{v_o}{R} \right) (-\hat{k}) + Mv_o R = \frac{7}{5} Mv_o R (-\hat{k})$$



8.5 Eccentric Impact

In eccentric impact the line of impact which is the common normal drawn at the point of impact does not pass through mass center of at least one of the colliding bodies. It involves change in state of rotation motion of either or both the bodies.

Consider impact of two A and B such that the mass center C_B of B does not lie on the line of impact as shown in figure. If we assume bodies to be frictionless their mutual forces must act along the line of impact. The reaction force of A on B does not pass through the mass center of B as a result state of rotation motion of B changes during the impact.



Problems of Eccentric Impact

Problems of eccentric impact can be divided into two categories. In one category both the bodies under going eccentric impact are free to move. No external force act on either of them. Their mutual forces are responsible for change in their momentum and angular momentum. In another category either or both of the bodies are hinged.

Eccentric Impact of bodies free to move

Since no external force acts on the two body system, we can use principle of conservation of linear momentum, principle of conservation of angular momentum about any point and concept of coefficient of restitution.

The coefficient of restitution is defined for components of velocities of points of contacts of the bodies along the line of impact.

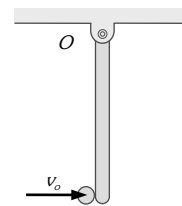
While applying principle of conservation of angular momentum care must be taken in selecting the point about which we write the equation. The point about which we write angular momentum must be at rest relative to the selected inertial reference frame and as far as possible its location should be selected on line of velocity of the mass center in order to make zero the first term involving moment of momentum of mass center.

Eccentric Impact of hinged bodies

When either or both of the bodies are hinged the reaction of the hinge during the impact act as external force on the two body system, therefore linear momentum no longer remain conserved and we cannot apply principle of conservation of linear momentum. When both the bodies are hinged we cannot also apply conservation of angular momentum, and we have to use impulse momentum principle on both the bodies separately in addition to making use of coefficient of restitution. But when one of the bodies is hinged and other one is free to move, we can apply conservation of angular momentum about the hinge.

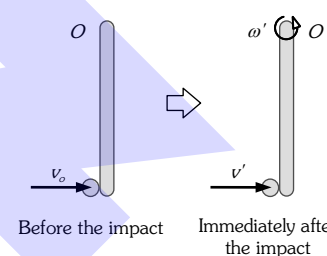
Illustrations

Illustration 60. A uniform rod of mass M and length ℓ is suspended from a fixed support and can rotate freely in the vertical plane. A small ball of mass m moving horizontally with velocity v_o strikes elastically the lower end of the rod as shown in the figure. Find the angular velocity of the rod and velocity of the ball immediately after the impact.



Solution.

The rod is hinged and the ball is free to move. External forces acting on the rod ball system are their weights and reaction from the hinge. Weight of the ball as well as the rod are finite and contribute negligible impulse during the impact, but impulse of reaction of the hinge during impact is considerable and cannot be neglected. Obviously linear momentum of the system is not conserved. The angular impulse of the reaction of hinge about the hinge is zero. Therefore angular momentum of the system about the hinge is conserved. Let velocity of the ball after the impact becomes v'_B and angular velocity of the rod becomes ω' .



We denote angular momentum of the ball and the rod about the hinge before the impact by L_{B1} and L_{R1} and after the impact by L_{B2} and L_{R2} .

Applying conservation of angular momentum about the hinge, we have

$$\vec{L}_{B1} + \vec{L}_{R1} = \vec{L}_{B2} + \vec{L}_{R2} \rightarrow mv_o \ell + 0 = mv'_B \ell + I_o \omega'$$

$$\text{Substituting } \frac{1}{3}M\ell^2 \text{ for } I_o, \text{ we have } 3mv'_B + M\ell\omega' = 3mv_o \quad \dots(1)$$

The velocity of the lower end of the rod before the impact was zero and immediately after the impact it becomes $\ell\omega'$ towards right. Employing these facts we can express the coefficient of restitution according to eq.

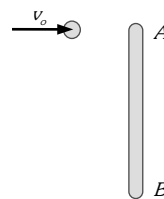
$$e = \frac{v'_{Qn} - v'_{Pn}}{v_{pn} - v_{Qn}} \rightarrow \ell\omega' - v'_B = ev_o \quad \dots(2)$$

From eq. (1) and (2), we have

$$\text{Velocity of the ball immediately after the impact } v'_B = \frac{(3m - eM)v_o}{3m + M}$$

$$\text{Angular velocity of the rod immediately after the impact } \omega' = \frac{3(1 + e)mv_o}{(3m + M)\ell}$$

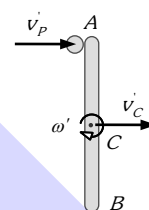
Illustration 61. A uniform rod AB of mass M and length ℓ is kept at rest on a smooth horizontal plane. A particle P of mass m_0 moving perpendicular to the rod with velocity v_0 strikes the rod at one of its ends as shown in the figure. Derive suitable expressions for the coefficient of restitution, velocity of mass center of the rod and angular velocity of the rod immediately after the impact. Assume e is the coefficient of restitution.



Solution.

Both the bodies can move freely in the horizontal plane, therefore no horizontal external force acts on the particle-rod system. The linear momentum as well as angular momentum about any axis normal to the plane is conserved.

Let the velocity of the particle, angular velocity of the rod and velocity of the mass center of the rod immediately after the impact are v'_p towards right, ω' in clockwise sense and v'_C towards right as shown in the adjacent figure. Using relative motion equation, we can express the velocity of the end A of the rod.



$$v'_A = v'_C + \frac{1}{2}\omega'\ell \quad (1)$$

We denote linear momentum of the particle and rod before the impact by $\vec{p}_{P1}$ and $\vec{p}_{C1}$ and immediately after the impact by $\vec{p}_{P2}$ and $\vec{p}_{C2}$ respectively.

Applying conservation of linear momentum, we have

$$\vec{p}_{P2} + \vec{p}_{C2} = \vec{p}_{P1} + \vec{p}_{C1} \rightarrow mv'_P + Mv'_C = mv_0 \quad (2)$$

The above equation shows that the mass center of the rod will move toward the right. If we write angular momentum of the rod about a stationary point O, which is in line with the velocity v'_C , the first term involving moment of momentum of rod vanishes and only angular momentum due to its centroidal rotation remains in the expression.

We denote angular momentum of the particle and the rod about the point O before the impact by L_{P1} and L_{R1} and after the impact by L_{P2} and L_{R2} .

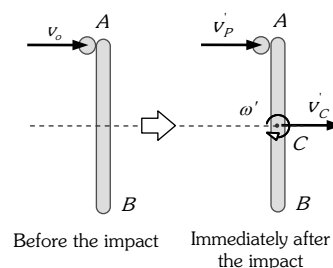
Applying conservation of angular momentum about the hinge, we have

$$\vec{L}_{P2} + \vec{L}_{R2} = \vec{L}_{P1} + \vec{L}_{R1} \rightarrow \frac{1}{2}mv'_P\ell + I_C\omega' = \frac{1}{2}mv_0\ell$$

Substituting $\frac{1}{12}M\ell^2$ for I_C , we have

$$6mv'_P + M\ell\omega' = 6mv_0 \quad (3)$$

The velocity of the end A of the rod before the impact was zero and immediately after the impact it becomes v'_A towards right.



Employing these facts we can express the coefficient of restitution as

$$e = \frac{v'_{Qn} - v'_{Pn}}{v_{pn} - v_{Qn}} \rightarrow v'_A - v'_P = ev_o$$

Substituting v'_A from eq. (1), we have

$$2v'_C + \omega' \ell - 2v'_P = 2ev_o \quad (4)$$

Eq. (2), (3) and (4) involves three unknowns v'_C , ω' and v'_P , which can be obtained by solving these equation.

Velocity of the ball immediately after the impact $v'_P = \left(\frac{4m - eM}{4m + M} \right) v_o$

Velocity of mass center of rod immediately after the impact $v'_C = \left\{ \frac{m(1+e)}{4m+M} \right\} v_o$

Angular velocity of the rod immediately after the impact $\omega' = \left\{ \frac{6m(1+e)}{4m+M} \right\} \frac{v_o}{\ell}$